

AKADEMIKA

PL-ARM7

ARM7 DEVELOPMENT KIT

EXPERIMENTAL MANUAL

WELCOME

Introduction

Today's system designers are faced with tomorrow's problems. EMBEDDED SYSTEM is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

Company's Motto

- Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

Important Information

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service center for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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TECHNICAL SPECIFICATIONS

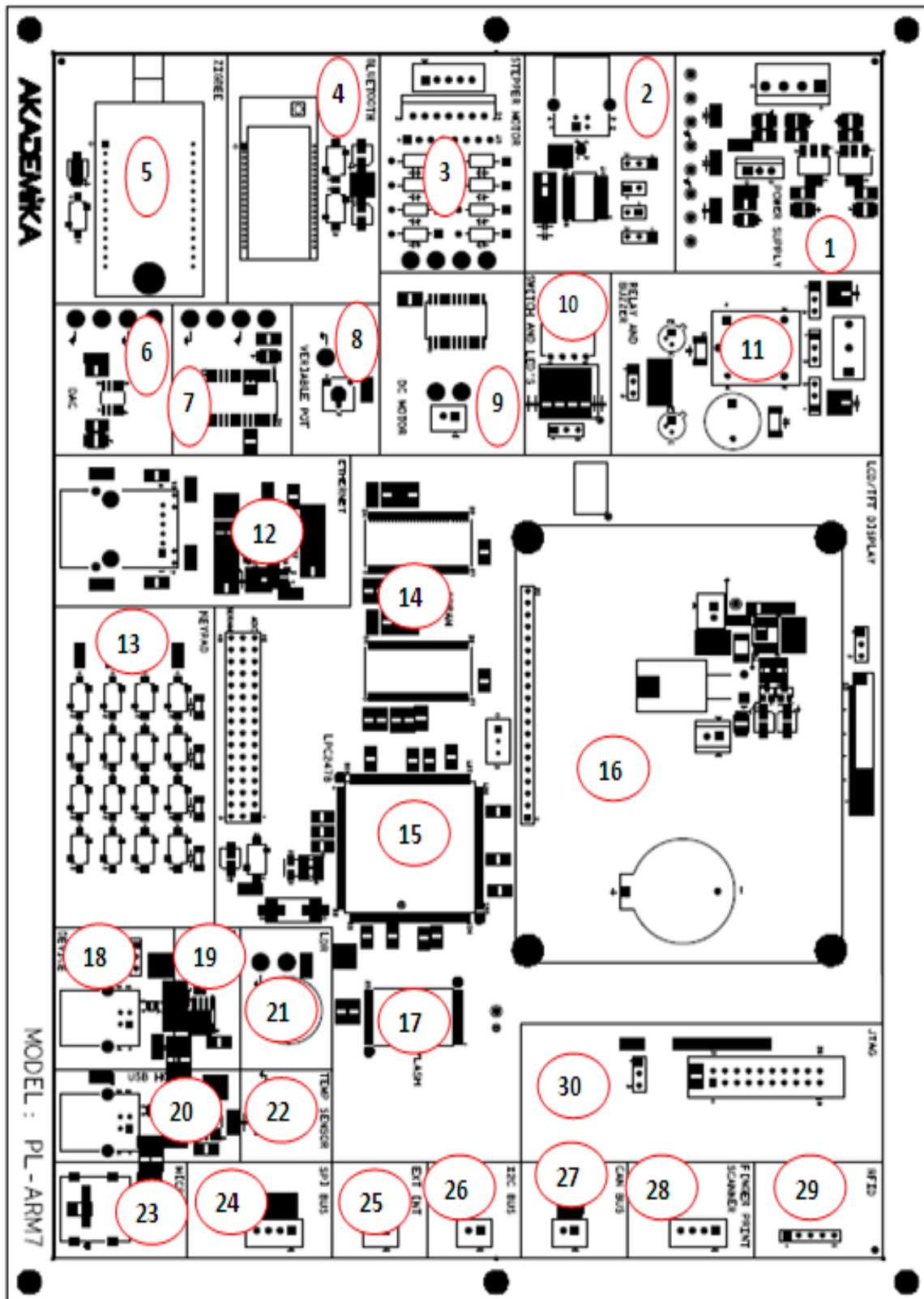
- | | |
|---|--|
| 1. Processor | : LPC2478. |
| 2. Operating frequency | : 72MHz. |
| 3. DISPLAY | :128X64 character Display |
| 4. Ethernet connectivity | : 10/100 MBPS Ethernet. |
| 5. USB | : USB Host & Device Connectivity. |
| 6. Programming | : ISP programming mode through USB. |
| 7. SENSORS | : Temp & Light sensor. |
| 8. Storage | : 16bit 2MB flash
: 512KB EEPROM |
| 9. ADC | : 12 bit Parallel ADC & internal 8bit ADC. |
| 10.DAC | : 16bit serial DAC & internal 8bit DAC. |
| 11.Motors interface | : 12V, 600mA, 1.8° stepper motor.
: 5 V DC motor. |
| 12.Relay & Buzzer | : 5V relay & Buzzer. |
| 13.General purpose switches & LEDs | : 4 switches & 4 LEDs. |
| 14.Keypad | : 4x4 keypad matrix. |
| 15.SD CARD | : 4GB SanDisk Memory card. |
| 16.Hardware protocol buses | : I2C, SPI, CAN bus. |

PL-ARM7 Deliverables

- **Accessory**

1. PL-ARM7 Development Board.
2. +5V, GND, +12V, GND power supply.
3. Power cord.
4. Shielded USB A to B cable.
5. MX1960 ADAPTOR.
6. 1 meter straight RJ45 LAN cable.
7. 30cm 2 pin Relimate cable.
8. 10 JUMPER CAPS.
9. 4 short links.
10. Experimental Manual.
11. Software CD
12. 12V, 600mA, 1.8° stepper motor.
13. 12V DC motor

FUNCTIONAL BLOCK



DESCRIPTION

1. Power supply section:-

Connect provided power supply to this section. In this section +12V supply is used for stepper motor. +3.3V is generated from +5V. +3.3V is used for LPC2478 & +5V used for peripherals.

2. ISP:-

USB to serial convertor used for programming to LPC2478 in ISP mode. In this section J63 & J64 selects LPC2478 TX/RX connection to connect to USB to serial or serial peripherals i.e. RFID, BLUETOOTH, ZIBBEE & Finger print scanner. J9 & J10 is to be connecting while downloading program in LPC2478 through flash magic.

3. Stepper Motor driver:-

This is stepper motor driver section; connect motor 5 pin connector to J14.

4. DAC section:-

In this section external & internal DAC is implemented. Output of external DAC is present on J70 with respect to J71 GND. Output of internal DAC is present on J55 with respect to J56 GND.

5. ADC section:-

In this section external & internal ADC is implemented. Output of external ADC is present on J68 with respect to J54 GND. Output of internal ADC is present on J53 with respect to J69 GND.

External ADC lines are shared with SDRAM hence connect the U23 connector to ADC side with the help of 2 pin jumper caps.

6. Variable POT:-

Variable analog voltage for ADC input.

7. DC motor:-

Connect 12V DC motor to J19 or in-between J43 & J46.

8. General purpose switches & LEDs:-

In this section, 4 general purpose LEDs & 4 general purpose switches are given for extra application or logic get applications etc.

9. Relay & Buzzer section:-

In this section, 5V relay & 5V buzzer is provided to make mechanical application and buzzer to generate tone. User can applied external supply to relay and connect your application through J7 phoenix connector.

10. Ethernet connection:-

In this section, Ethernet PHY is connected to LPC2478 with RMII (Reduced Media Independent Interface). And PHY is connected to Ethernet connector.

11. Keypad Matrix:-

In this section, 16 keys are implemented in 4X4 matrix form. It can be used as a hex keypad.

12. LPC2478 section:-

This is the LPC2478 IC with is DCAPs. Hear 12MHz crystal oscillator is used as main oscillator and 32.7456 KHz crystal oscillator is used for internal battery Backup RTC.

13. LCD section:-

In this section, user can connect GLCD to J8 connector. At a time only one display will work.. Hear 128 X 64 dots GLCD are used.

14. Flash section:-

In this section, 16bit 2MB flash is used. This flash is command based flash. This flash is connected to LPC2478 through EMC peripheral at 0X80000000 address space.

15. USB device:-

This is USB device section, while running this program this interface act as mass storage device to the computer. While performing USB device experiment, connect USB to PC from J49 connector.

16. EEPROM section:-

In this section, we used 512KB I2C based EEPROM, to read write data. This EEPROM having 0XA0 and 0XA1 write & read address.

17. USB Host :-

This is USB Host section, while running this program this interface act as a computer to access mass storage devices. While performing USB device experiment, connect pen drive to J48 through USB male female cable.

18. LDR section:-

In this section light sensor is used for experiment.

19. Temperature sensor:-

In this experiment, LM35 temperature section is used for temperature reading application.

20. SD card section:-

In this section, we interface 4 GB micro SD card from SanDisk. SD card is interfaced in SPI mode.

21. SPI bus:-

This is SPI bus for your external hardware connection and visualizing SPI protocol by using logic analyzer.

22. External interrupt:-

In this section we can apply external signal to LPC2478 as an interrupt signal. This signal must in 0s and 1s format i.e. 0V and +3.3V.

23. I2C bus:-

This is I2C bus for your external hardware connection and visualizing I2C protocol by using logic analyzer.

24. CAN bus:-

In this section CAN bus is implemented. User can communicate CAN bus in self-test mode or communicate between two boards in TX/RX mode.

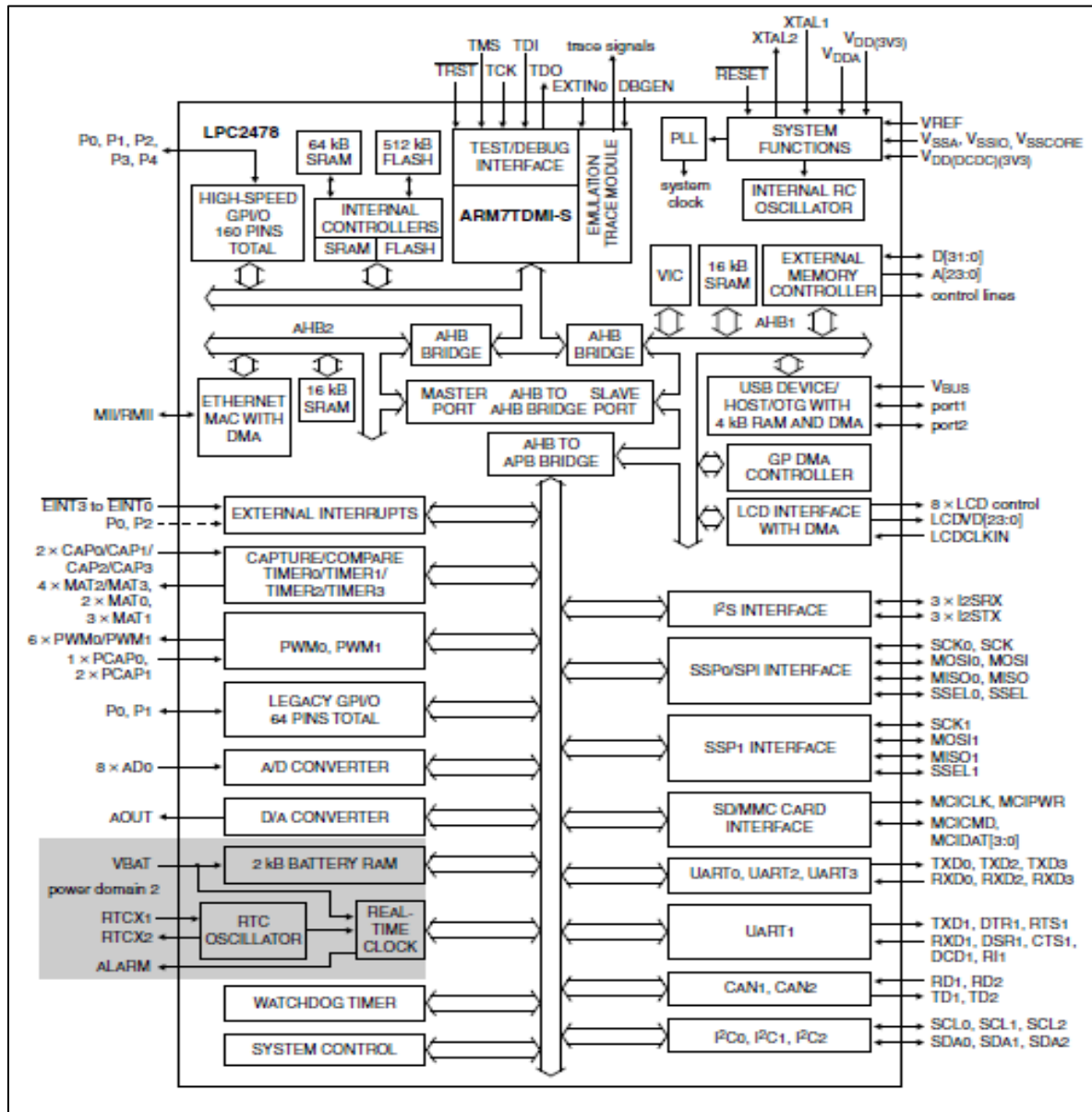
25. JTAG debugger interface.

In this section, JTAG connector is available for user to interface debugger.

INTRODUCTION

PL-ARM development board is designed towards the practical implementation of ARM7 interfacing with different peripherals like Graphics display interfacing, Flash, EEPROM interface, USB HOST, USB DEVICE interface, Ethernet interface, DC, Stepper motor interface, ADC, DAC, SD Card, Hardware protocol like I2C, SPI & CAN bus, Relay, buzzer, switch, LEDs etc.

LPC2478 block diagram



INTRODUCTION TO LPC2478

NXP Semiconductor designed the LPC2400 microcontrollers around a 16-bit/32-bit

ARM7TDMI-S CPU core with real-time debug interfaces that include both JTAG and embedded Trace. The LPC2400 microcontrollers have 512 kb of on-chip high-speed

Flash memory. This Flash memory includes a special 128-bit wide memory interface and accelerator architecture that enables the CPU to execute sequential instructions from Flash memory at the maximum 72 MHz system clock rate. This feature is available only on the LPC2000 ARM Microcontroller family of products.

The LPC2400 can execute both 32-bit ARM and 16-bit Thumb instructions.

Support for the two Instruction Sets means Engineers can choose to optimize their application for either performance or code size at the sub-routine level.

When the core executes instructions in Thumb state it can reduce code size by more than 30 % with only a small loss in performance while executing instructions in ARM state maximizes core performance.

The LPC2400 microcontrollers are ideal for multi-purpose communication applications. It incorporates a 10/100 Ethernet Media Access Controller (MAC), a USB full speed device/host/OTG controller with 4 kb of endpoint RAM, four UARTs, two Controller Area Network (CAN) channels, an SPI interface, two Synchronous Serial Ports (SSP), three I2C interfaces, and an I2S interface.

Supporting this collection of serial communications interfaces are the following feature components; an on-chip 4 MHz internal precision oscillator, 98 kb of total RAM consisting of 64 kb of local SRAM, 16 kb SRAM for Ethernet, 16 kb SRAM for general purpose DMA, 2 kb of battery powered SRAM, and an External Memory Controller (EMC). These features make this device optimally suited for communication gateways and protocol converters. Complementing the many serial communication controllers, versatile clocking capabilities, and memory features are various 32-bit timers, an improved 10-bit ADC, 10-bit DAC, two PWM units, four external interrupt pins, and up to 160 fast GPIO lines. The LPC2400 connect 64 of the GPIO pins to the hardware based Vector Interrupt Controller (VIC) that means these external inputs can generate edge-triggered, interrupts. All of these features make the LPC2400 particularly suitable for industrial control and medical systems.

LPC2478 Features

- ARM7TDMI-S processor, running at up to 72 MHz
- 98 kb on-chip SRAM includes:
 - 64 kb of SRAM on the ARM local bus for high performance CPU access.
 - 16 kb SRAM for Ethernet interface. Can also be used as general purpose SRAM.
 - 16 kb SRAM for general purpose DMA use also accessible by the USB.

- 2 kb SRAM data storage powered from the RTC power domain.

- **LPC2458/68/78 only:**

512 kb on-chip Flash program memory with In-System Programming (ISP) and In-Application Programming (IAP) capabilities. Flash program memory is on the ARM local bus for high performance CPU access.

- Dual Advanced High-performance Bus (AHB) system allows memory access by multiple resources and simultaneous program execution with no contention.

- EMC provides support for asynchronous static memory devices such as RAM, ROM and Flash, as well as dynamic memories such as Single Data Rate SDRAM.

- Advanced Vectored Interrupt Controller (VIC), supporting up to 32 vectored interrupts.

- General Purpose AHB DMA controller (GPDMA) that can be used with the SSP, I2S, and SD/MM interface as well as for memory-to-memory transfers.

- **LPC2470/78 only:** LCD controller, supporting both Super-Twisted Nematic (STN) displays.

- Dedicated DMA controller.
- Selectable display resolution (up to 1024 × 768 pixels).
- Supports up to 24-bit true-color mode.

- **Serial Interfaces:**

- Ethernet MAC with MII/RMII interface and associated DMA controller. These functions reside on an independent AHB bus.
- USB 2.0 full-speed dual port device/host/OTG controller with on-chip PHY and associated DMA controller.
- Four UARTs with fractional baud rate generation, one with modem control I/O, and one with IrDA support, all with FIFO.
- CAN controller with two channels.
- SPI controller.
- Two SSP controllers, with FIFO and multi-protocol capabilities. One is an alternate for the SPI port, sharing its interrupt. SSPs can be used with the GPDMA controller.
- Three I2C-bus interfaces (one with open-drain and two with standard port pins).
- I2S (Inter-IC Sound) interface for digital audio input or output. It can be used with the GPDMA.

- **Other peripherals:**

- SD/MMC memory card interface.
- 160 general purpose I/O pins with configurable pull-up/down resistors.

- 10-bit ADC with input multiplexing among 8 pins.
 - 10-bit DAC.
 - Four general purpose timers/counters with 8 capture inputs and 10 compare outputs. Each timer block has an external count input.
 - Two PWM/timer blocks with support for three-phase motor control. Each PWM has an external count inputs.
 - Real-Time Clock (RTC) with separate power domain, clock source can be the RTC oscillator or the APB clock.
 - 2 kb SRAM powered from the RTC power pin, allowing data to be stored when the rest of the chip is powered off.
 - Watch Dog Timer (WDT). The WDT can be clocked from the internal RC oscillator ,the RTC oscillator, or the APB clock.
-
- Standard ARM test/debug interface for compatibility with existing tools.
 - Emulation trace module supports real-time trace.
 - Single 3.3 V power supply (3.0 V to 3.6 V).
 - Four reduced power modes: idle, sleep, power-down, and deep power-down.
 - Four external interrupt inputs configurable as edge/level sensitive. All pins on PORT0 and PORT2 can be used as edge sensitive interrupt sources.
 - Processor wake-up from Power-down mode via any interrupt able to operate during Power-down mode (includes external interrupts, RTC interrupt, USB activity, Ethernet wake-up interrupt, CAN bus activity, PORT0/2 pin interrupt).
 - Two independent power domains allow fine tuning of power consumption based on needed features.
 - Each peripheral has its own clock divider for further power saving. These dividers help reducing active power by 20 - 30 %.
 - Brownout detects with separate thresholds for interrupt and forced reset.
 - On-chip power-on reset.
 - On-chip crystal oscillator with an operating range of 1 MHz to 24 MHz
 - 4 MHz internal RC oscillator trimmed to 1 % accuracy that can optionally be used as the system clock. When used as the CPU clock, does not allow CAN and USB to run.

-
- On-chip PLL allows CPU operation up to the maximum CPU rate without the need for a high frequency crystal. May be run from the main oscillator, the internal RC oscillator, or the RTC oscillator.
 - Boundary scans for simplified board testing.
 - Versatile pin function selections allow more possibilities for using on-chip peripheral functions.

JUMPER DETAILS

1. J52, CAN BUS

1	CANL
2	CANH

2. JP15, I2C

1	SDA
2	SCL

3. JP65, EXT INT

1	I/P
2	GND

4. J16, SPI BUS

1	SCK
2	SSEL
3	MISO
4	MOSI

SOFTWARE INSTALLATION

1. Install IAR workbench for ARM as follows.

- Double click on



EWARM-KS-WEB-5502

- Following window will appear.

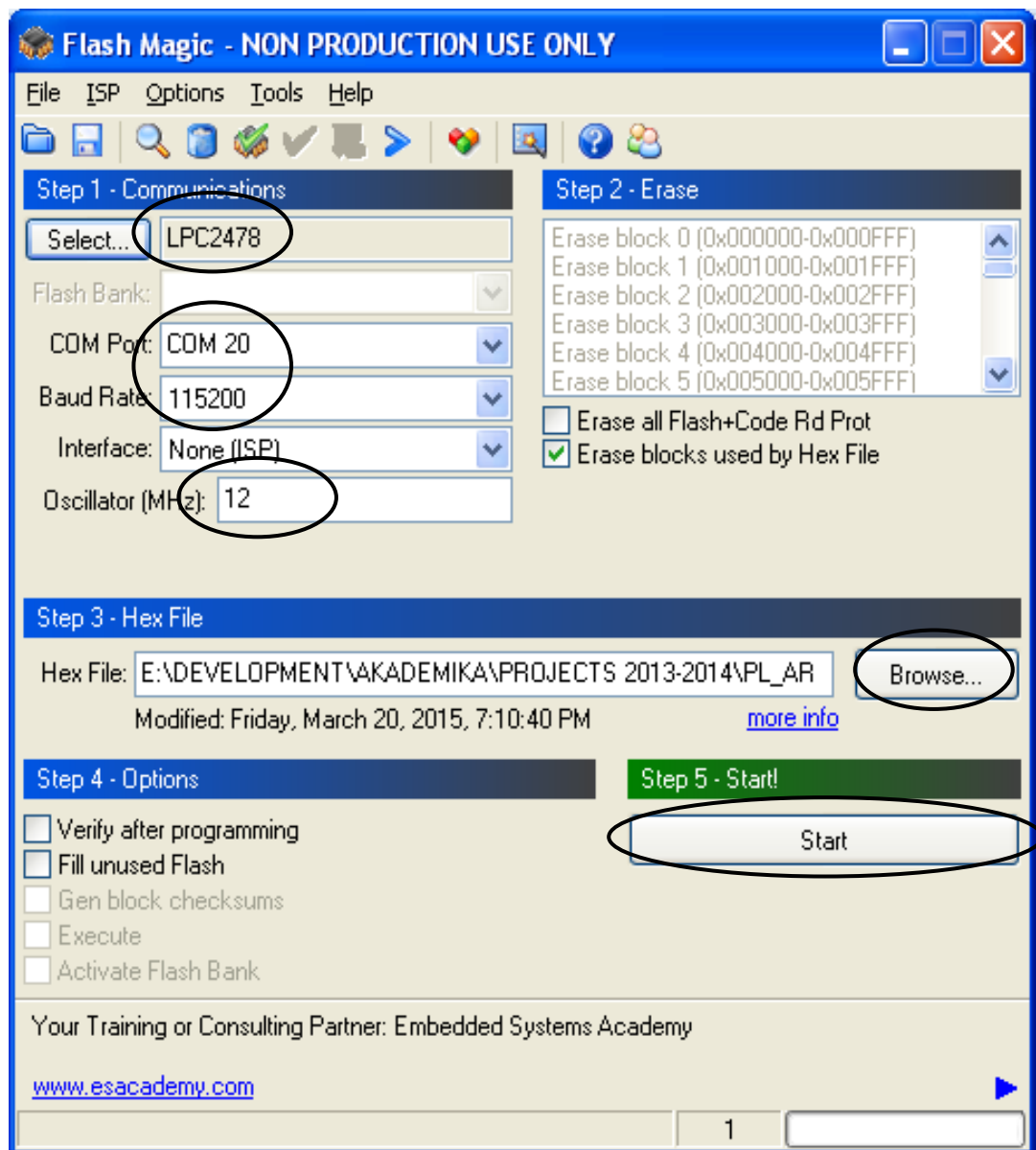


- Click on Install IAR Embedded workbench from above window.
- It ask for License No, then put **9546-651-909-9023**
- It ask for License key, then put
SXTO84BEYS23QCGQO0PXBCYQ3AJWB89LLZ7UVY8NK3S5N
DWCIYEPVKOIGI6EIXI4ONZ73ZTEUDN6XOD4L36ID8VO3J9UD78ZZ
UQO6A4YUDNVCQUTBPFNZRDXARUR5D1WQGMNIRSZHKOJT
VYAC5IB4T22FEKLHY12BOSYOH0PJAIHQQDSAGFMEZFRG
GQCSLOBW#

2. USB driver installation

- For programming, connect USB cable to J11 in ISP section.
- PC asks for driver then gives path of driver folder from CD provided by Akademika.
- For different operating system, different drivers are present in CD.
- See the comport which needed while programming.

3. Install flash magic software provided.
4. Flash magic setting, as follows



Select comport as per shown in device manger.
Browse hex file to download in LPC2478.
Click on Start to download program.

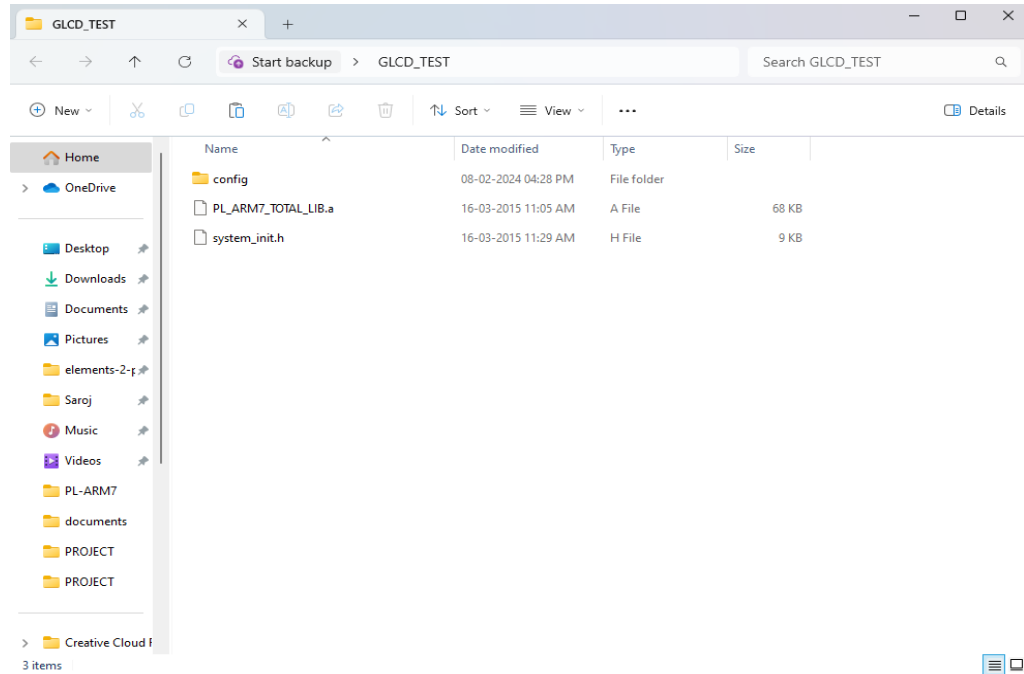
HOW TO CREATE A NEW PROJECT

1. First Create a folder with your Project name.

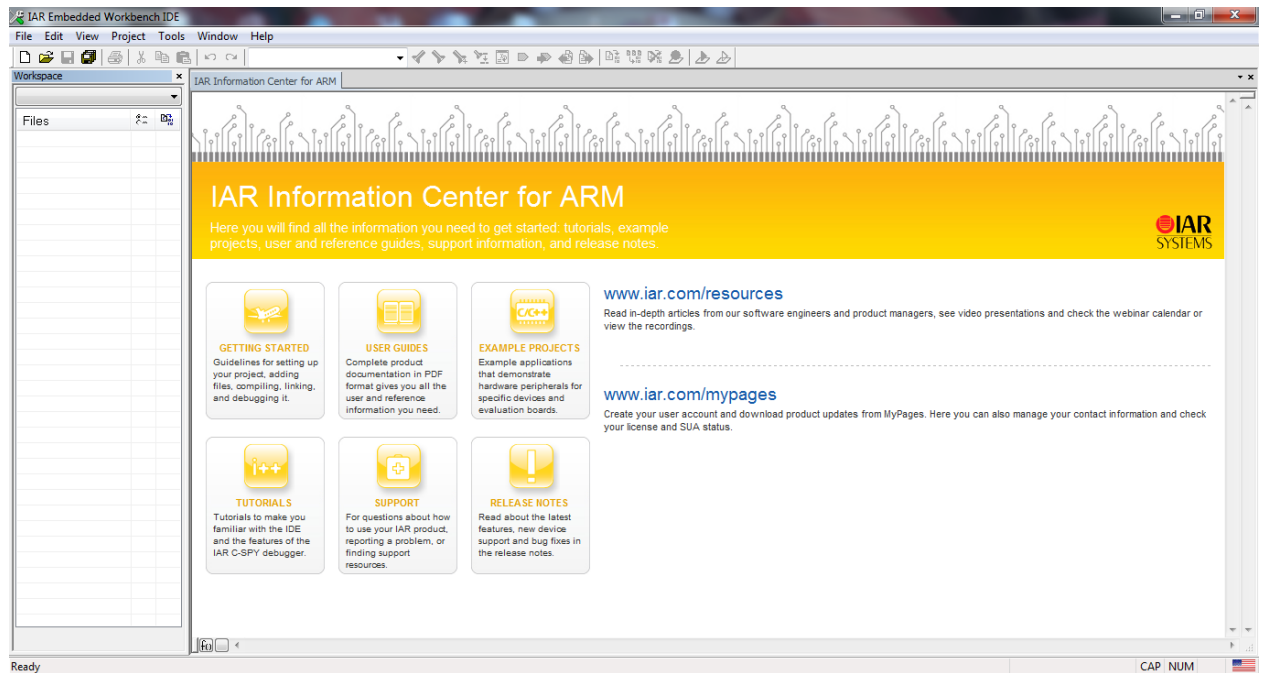
And copy this File :-

- Config
- PL_ARM7_TOTAL_LIB.a
- system_init.h.

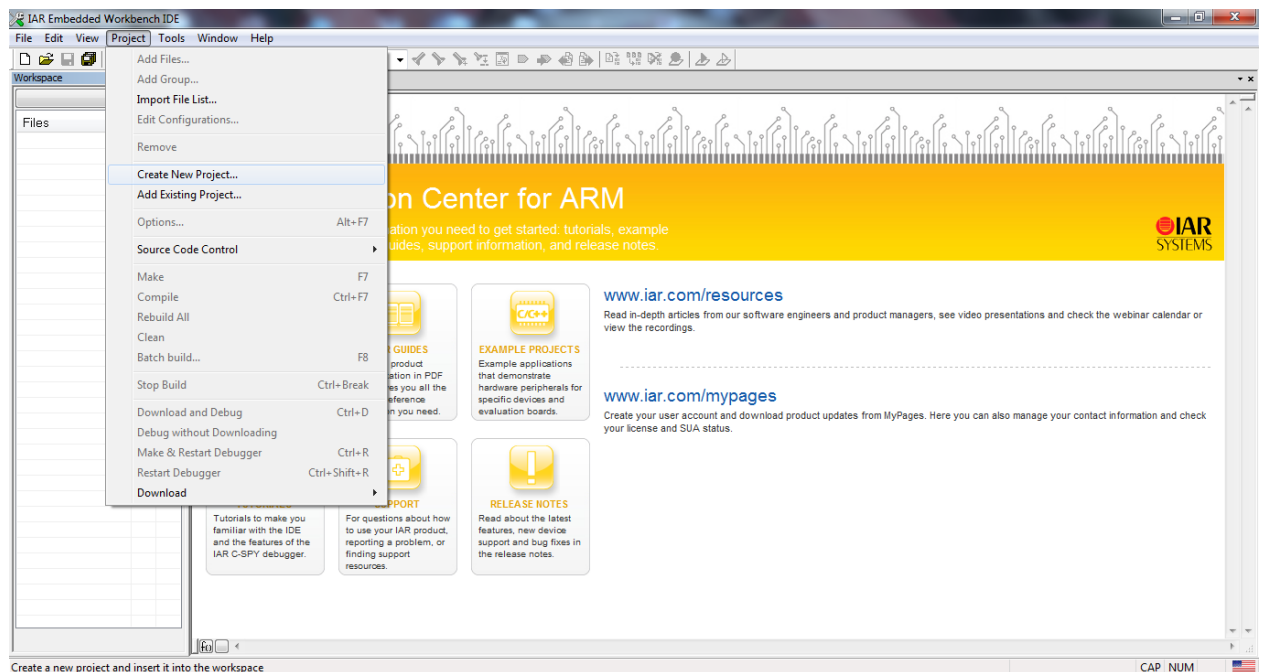
This Files Included in CD provided by Akademika..



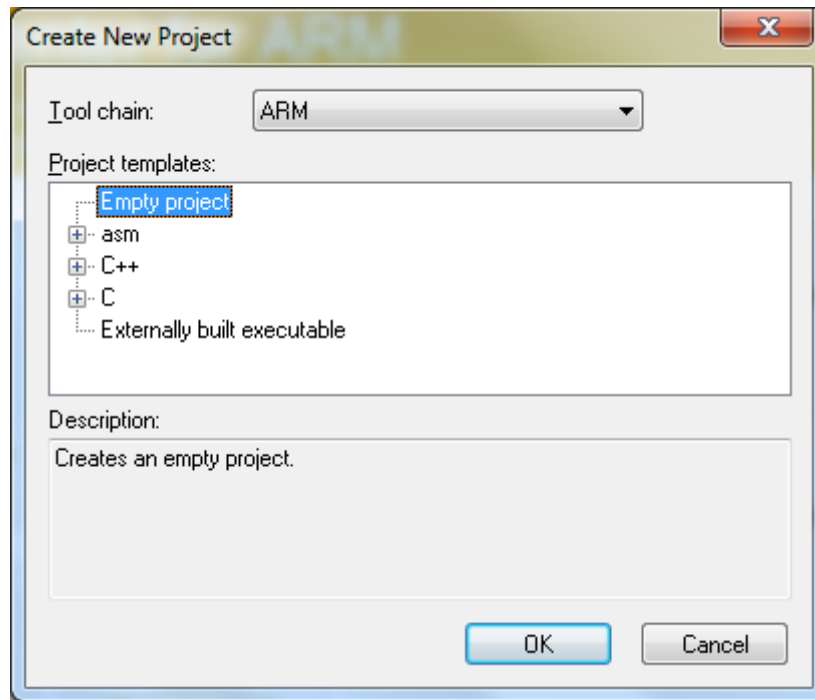
2. Open IAR Embedded Workbench IDE tool



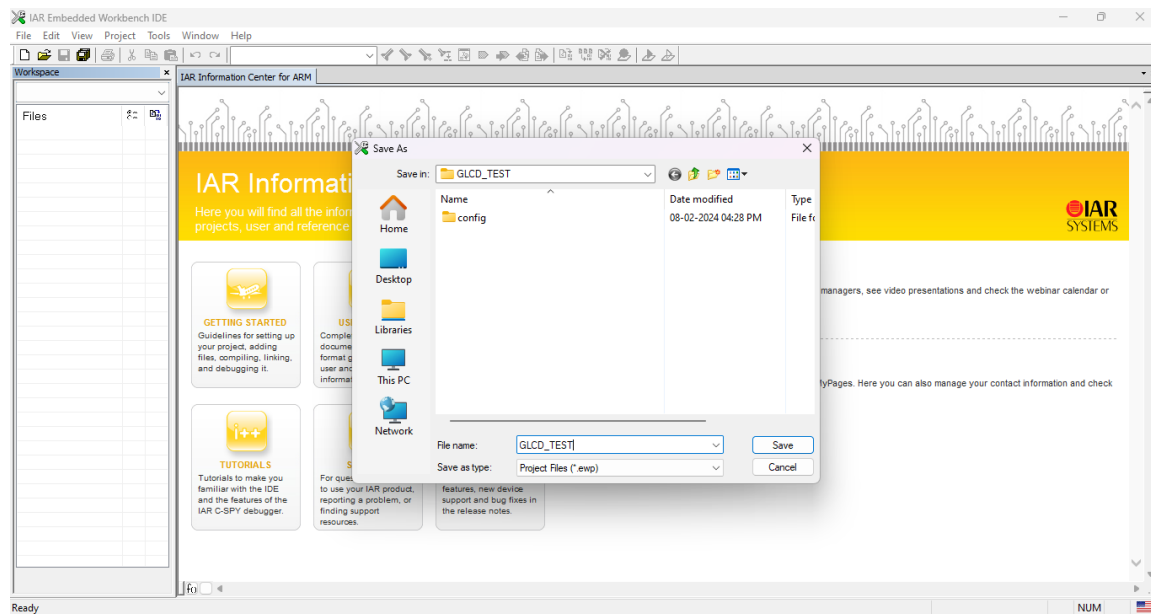
3. Go to Project -> Create New Project

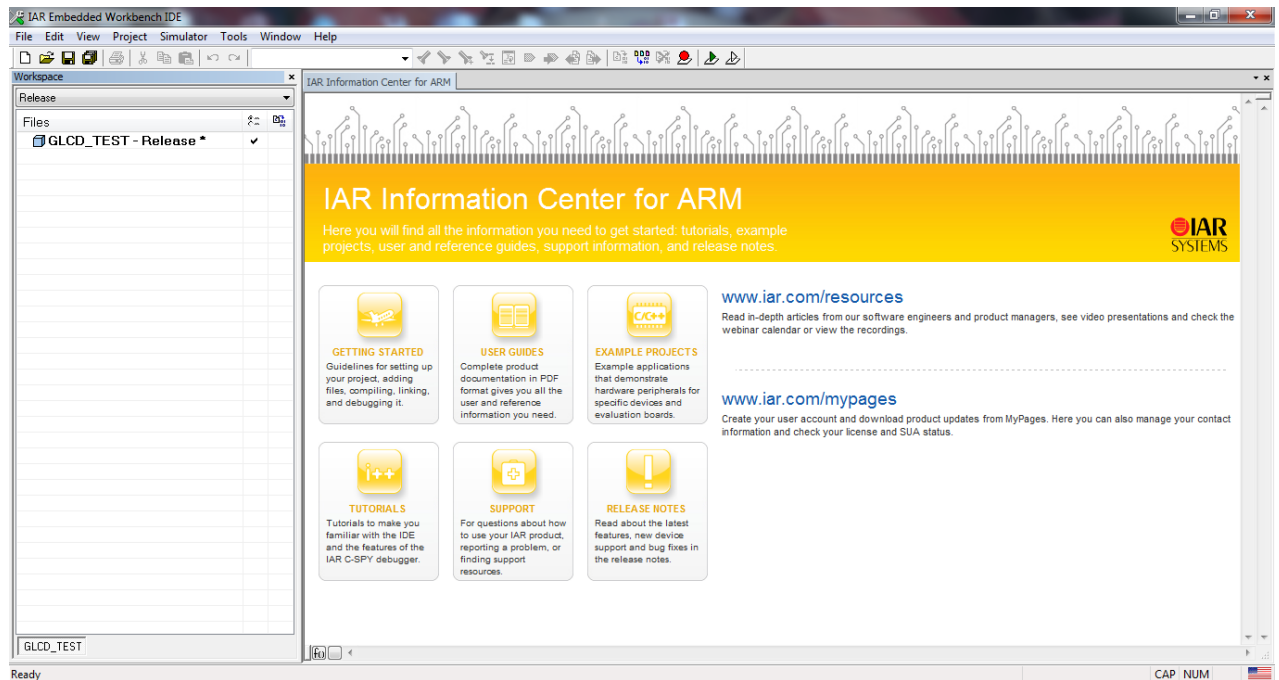


4. Select **Tool chain: ARM ->Empty Project ->OK**

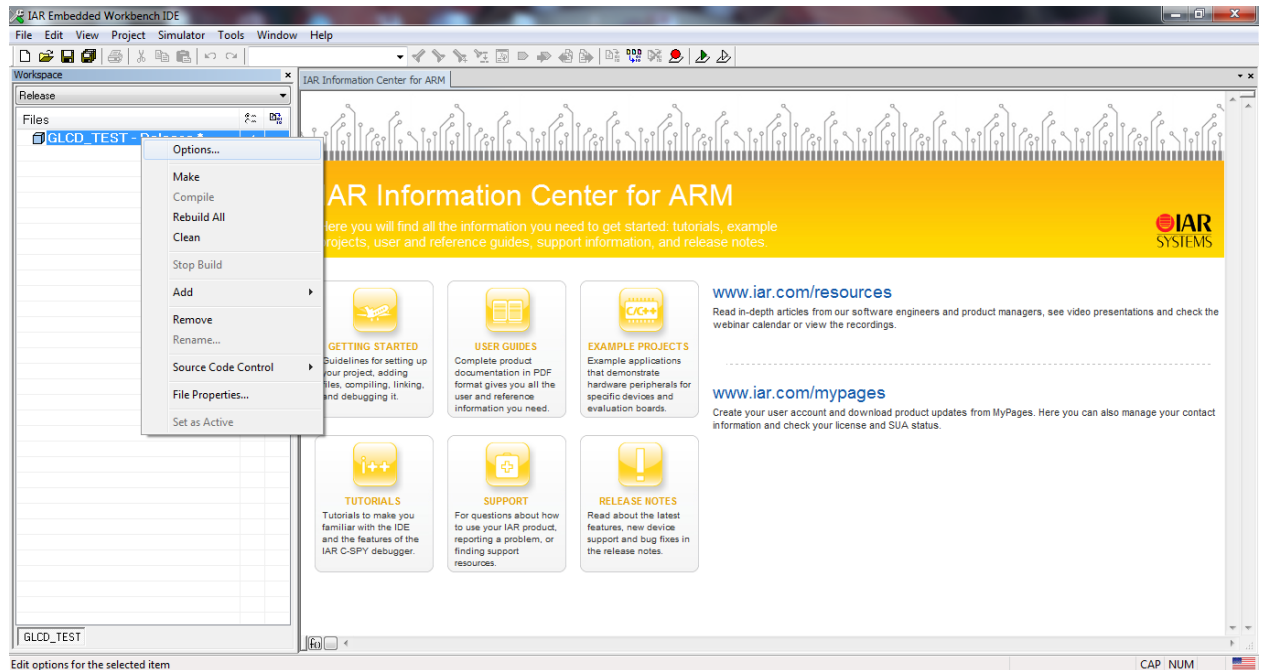


5. Type File name: **GLCD_TEST** -> Select Save as type: **Project Files** -> **>Save**

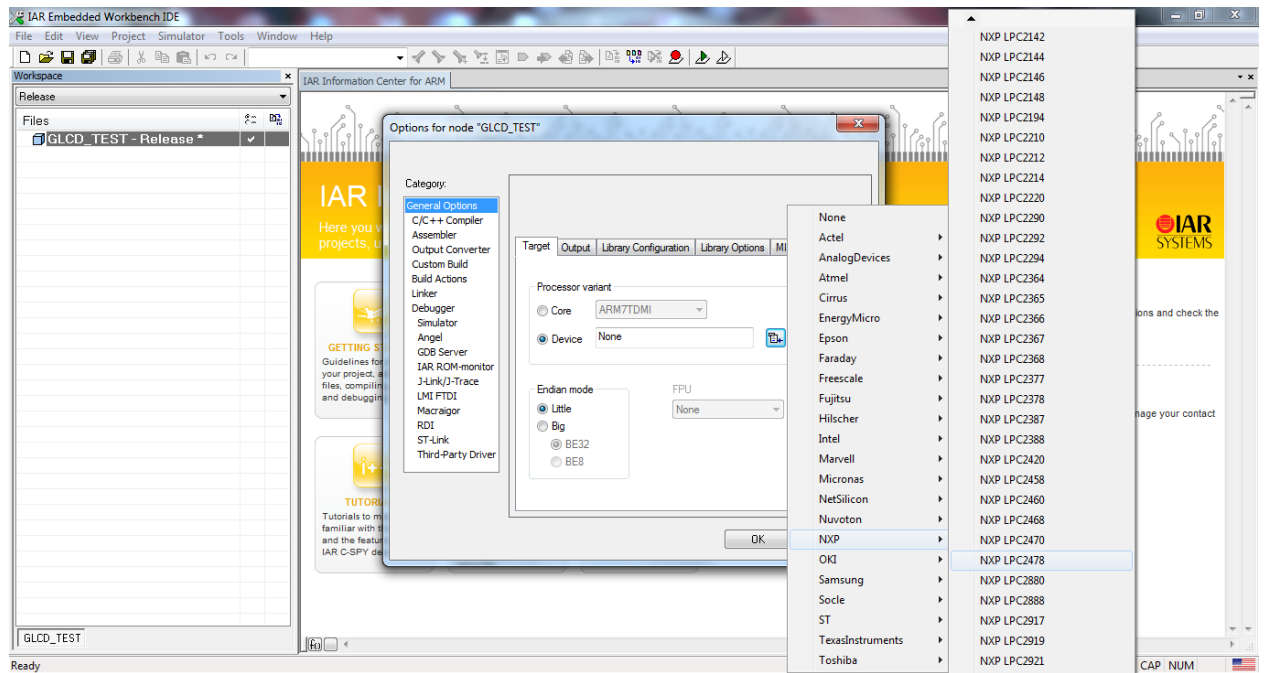




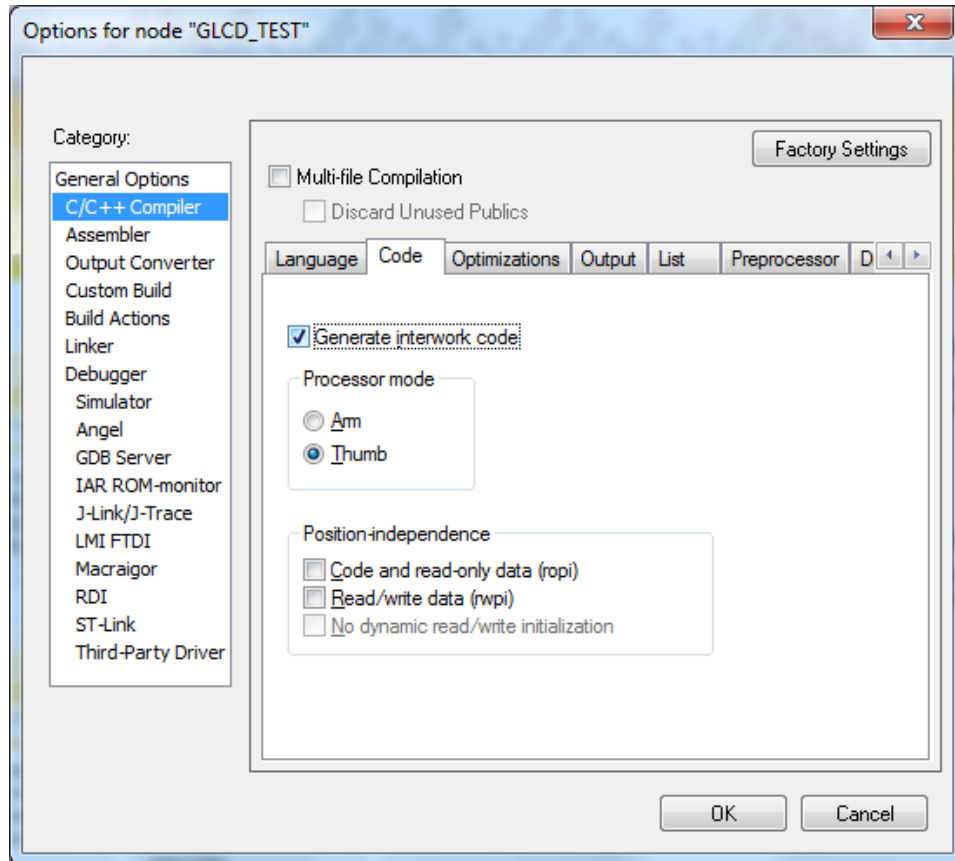
6. Right Click on **GLCD_TEST** -> Select Options



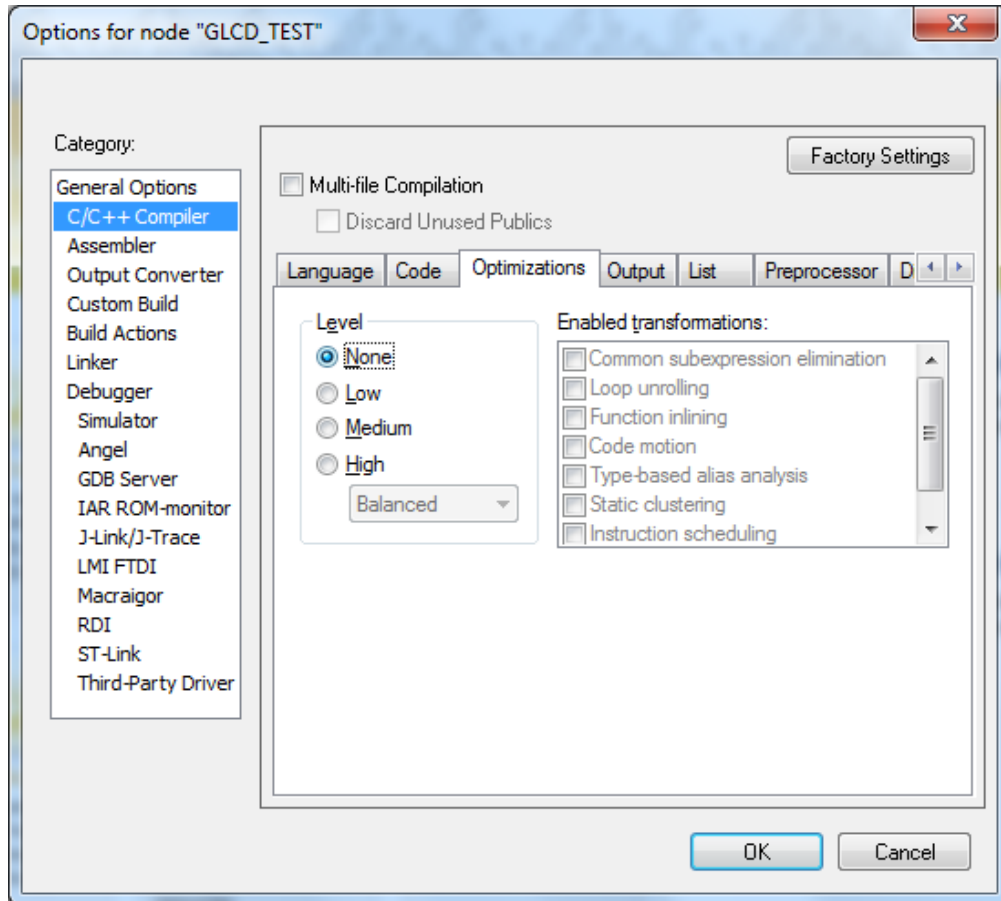
7. Under **Category** -> Select **General Options** -> **Target**, in Processor variant -> Select **Device** -> Select **NXP** -> Click on **NXP LPC2478** and in Endian mode -> Select **Little**.

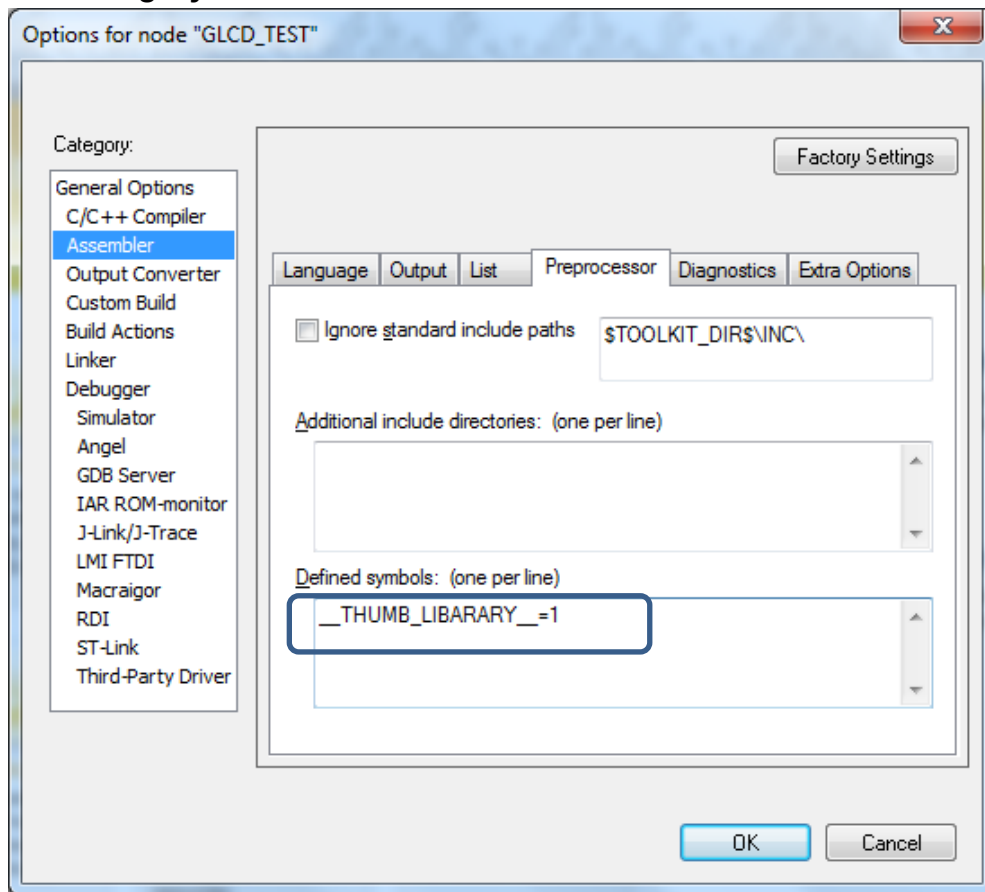


8. Under **Category** -> Select **C/C++ Compiler** -> **Code** -> Check **General interwork code** -> Select **Thumb**.

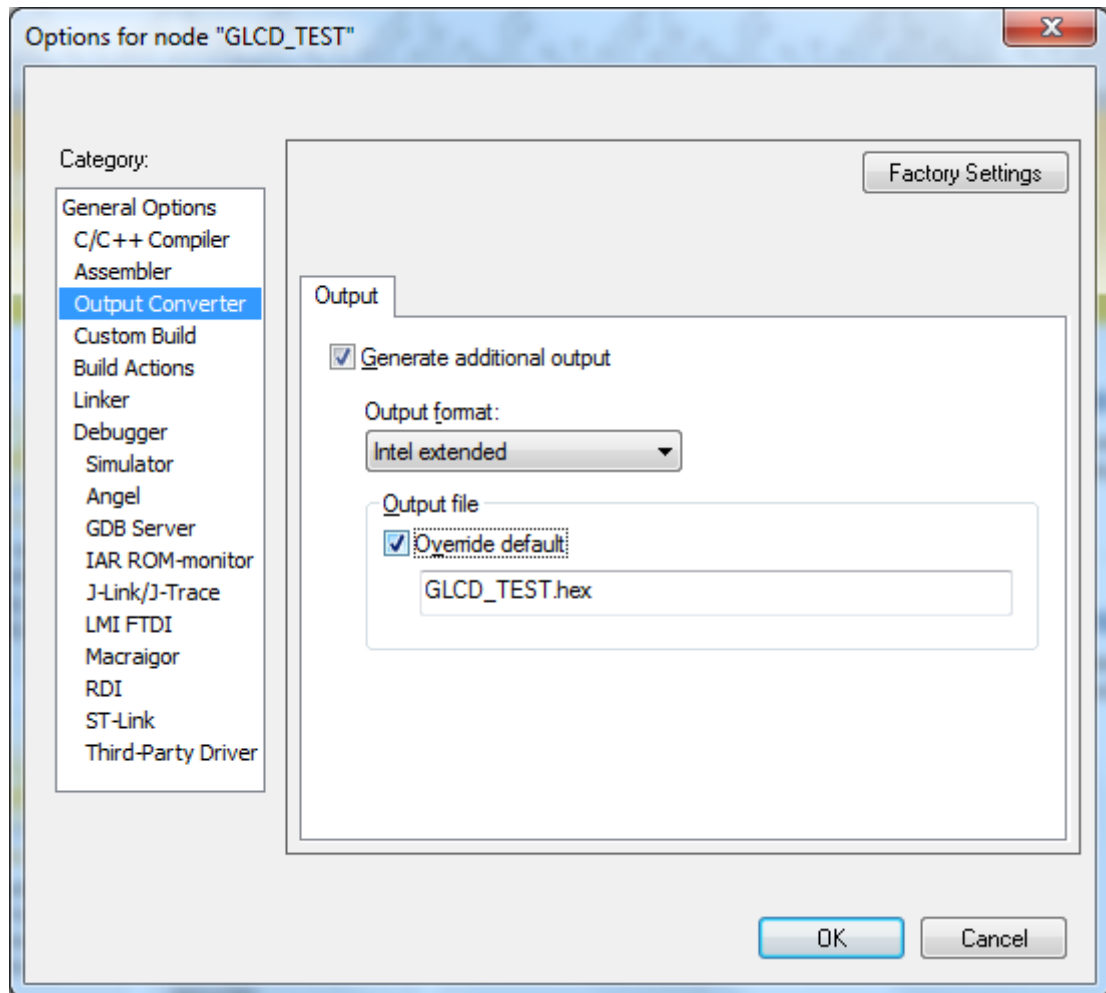


9. Under **Category** -> Select **C/C++ Compiler** -> **Optimizations** -> Select **None** -> **OK**

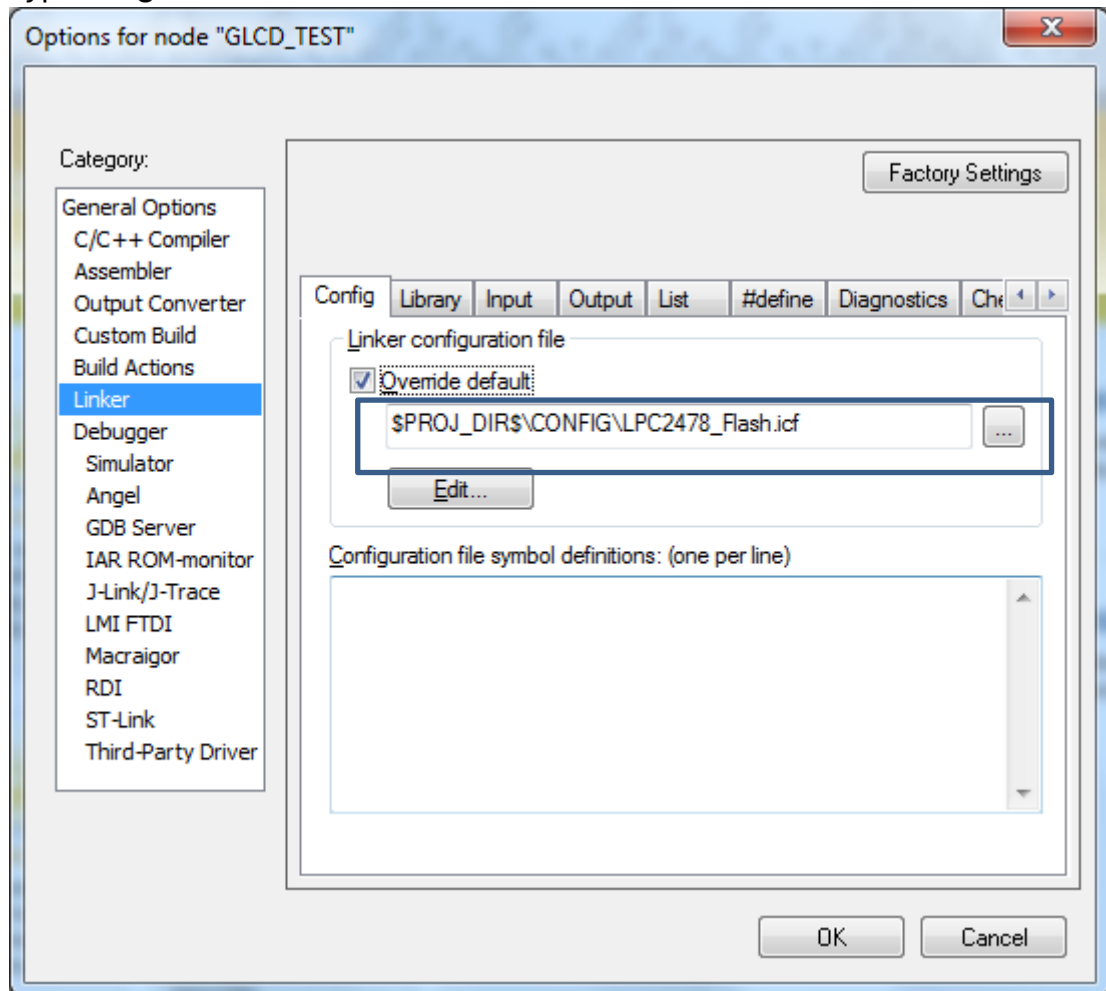


10. Under Category -> Select Assembler ->OK

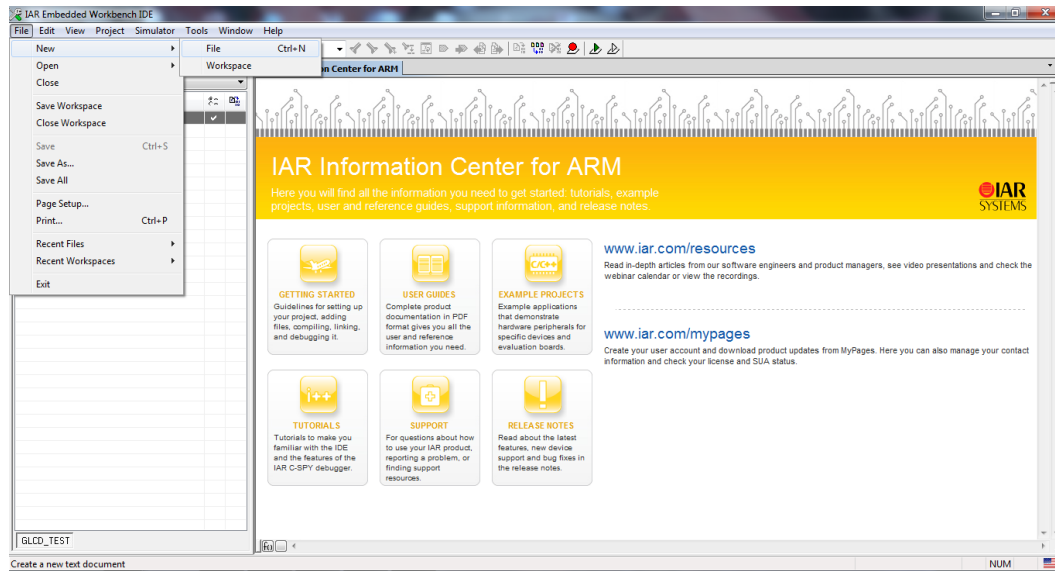
11. Under **Category** -> Select **Output Converter** -> Check **Generate additional output** -> Output format: Select **Intel extended** -> Check **Override default** ->OK



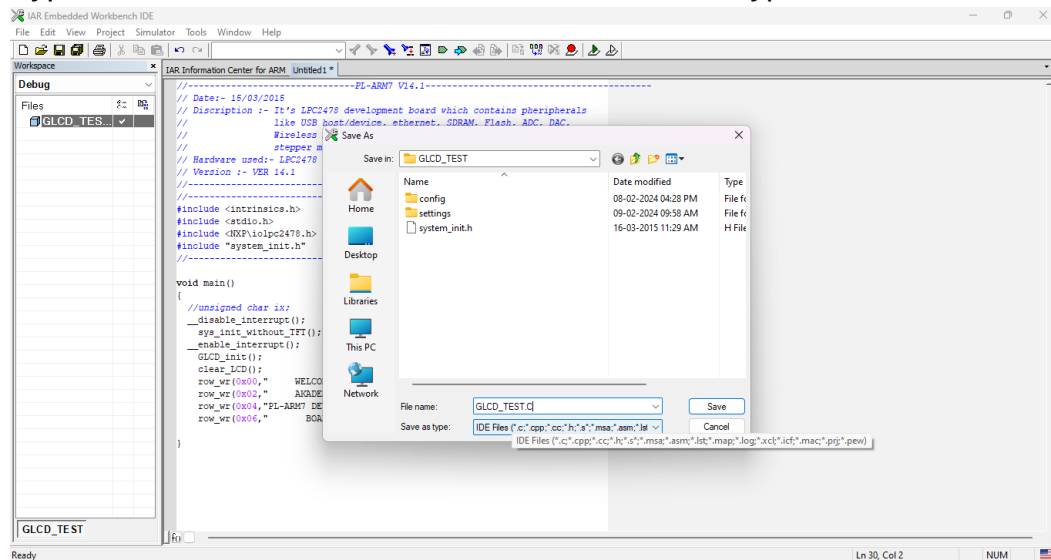
12. Under **Category** -> Select **Linker** -> **Config** -> Check **Override default** -> Type **Hilights Text** -> **OK**



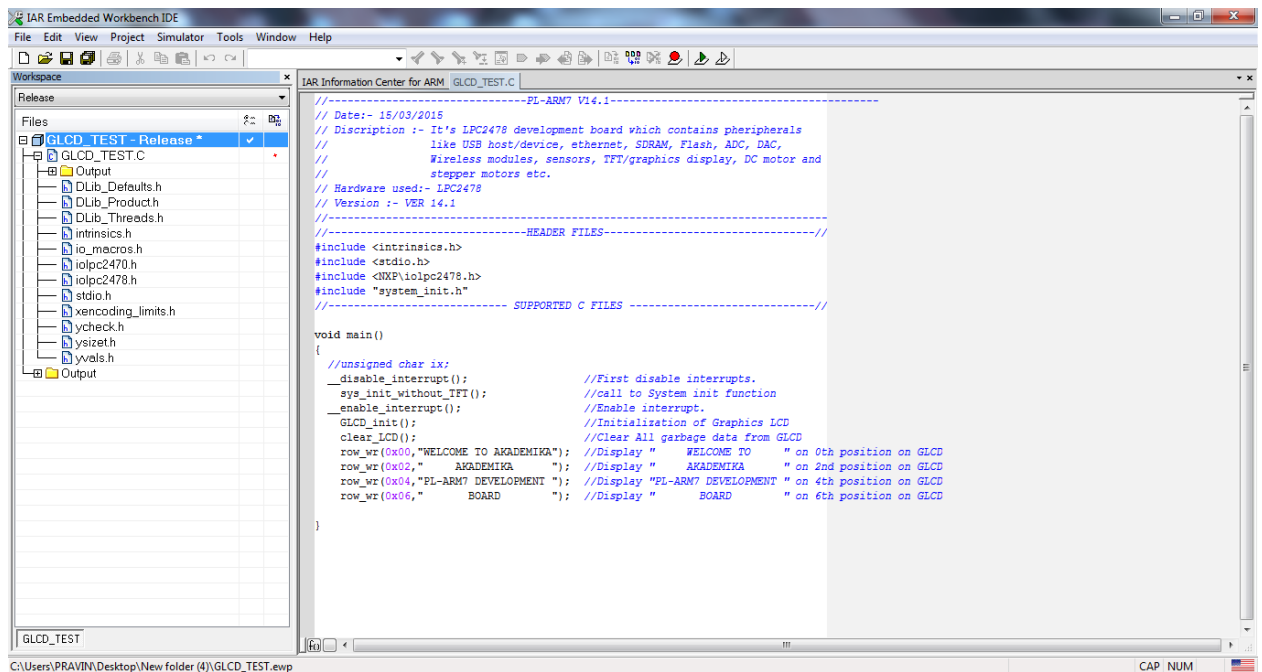
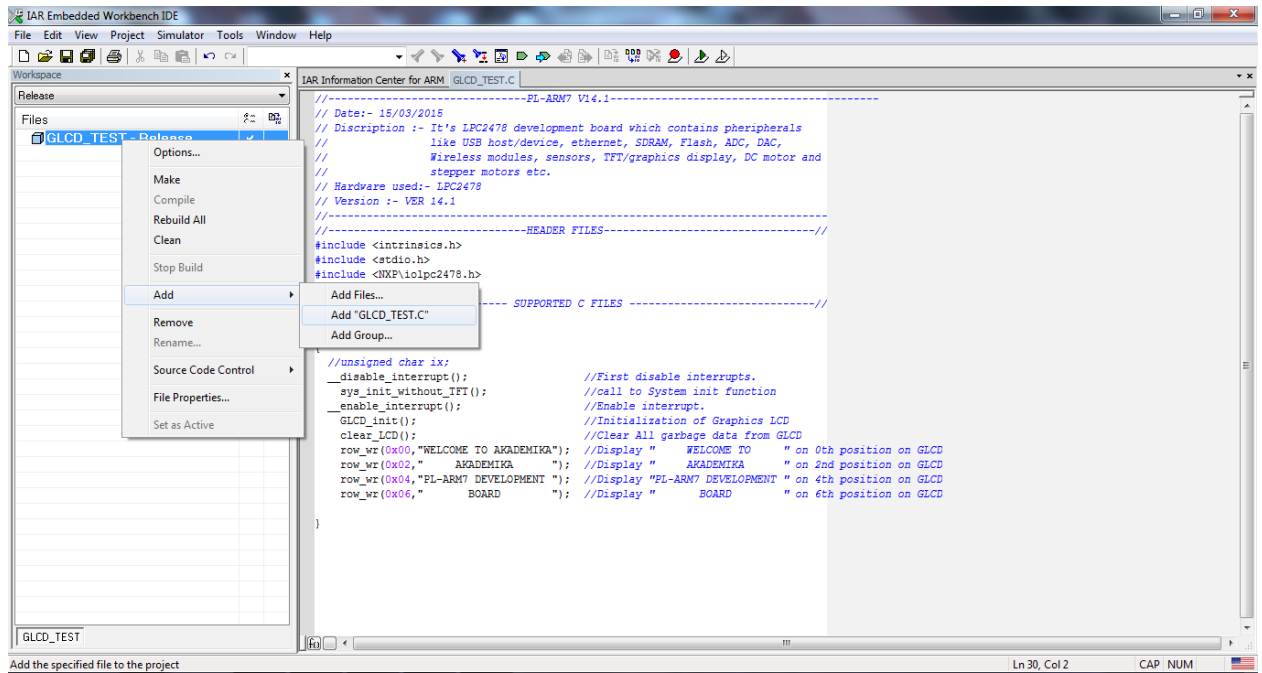
13. Go to **File ->New ->File**



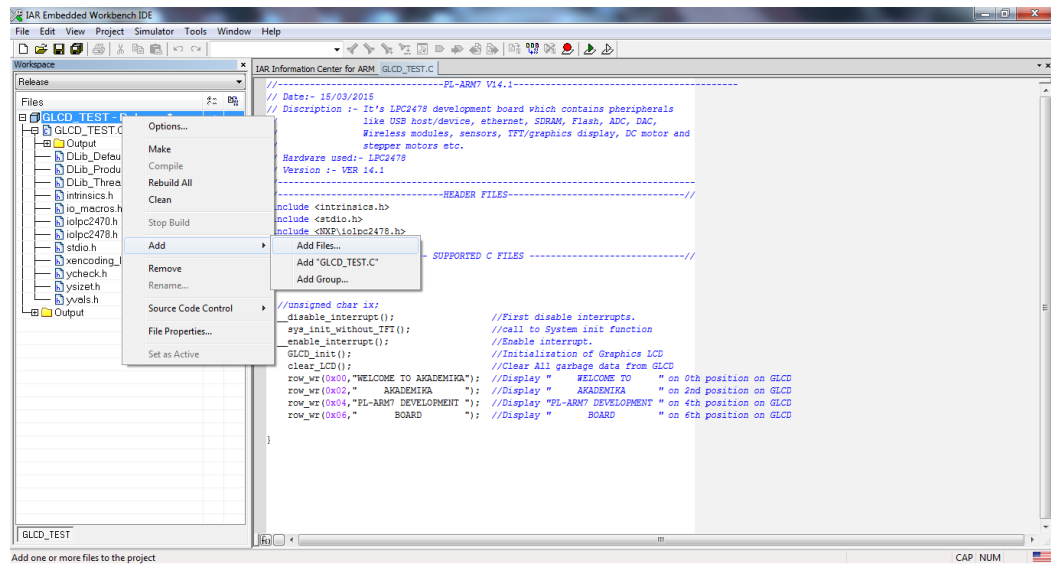
14. Type File name: **GLCD_TEST.C** -> Select Save as type: **IDE Files** ->**Save**



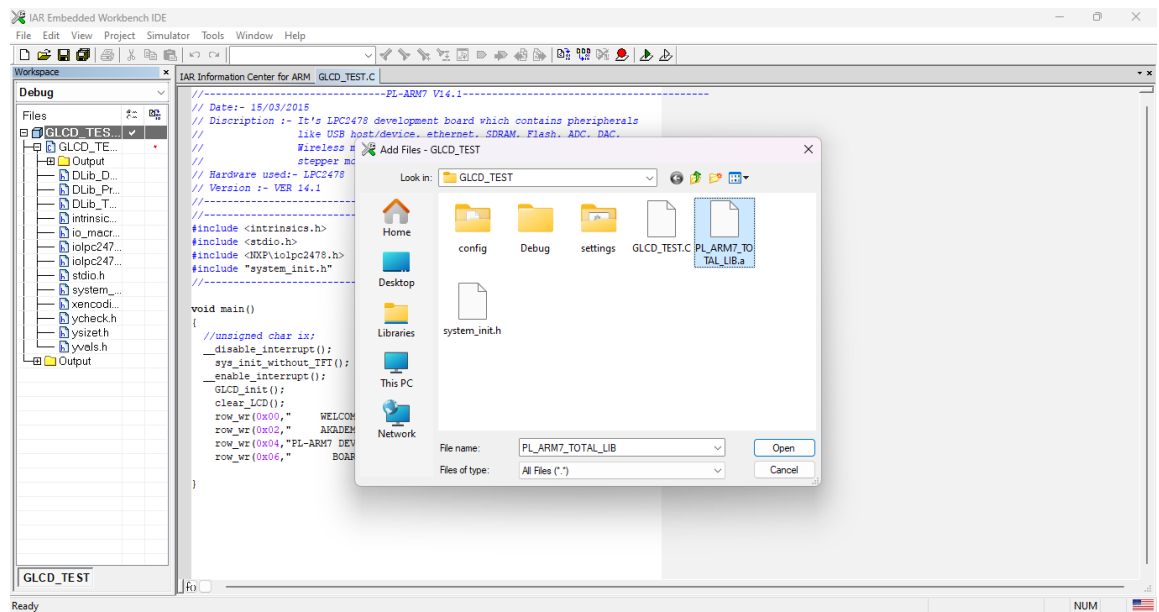
15. Right Click on **GLCD_TEST** -> **ADD** -> Select **Add "GLCD_TEST.C"**



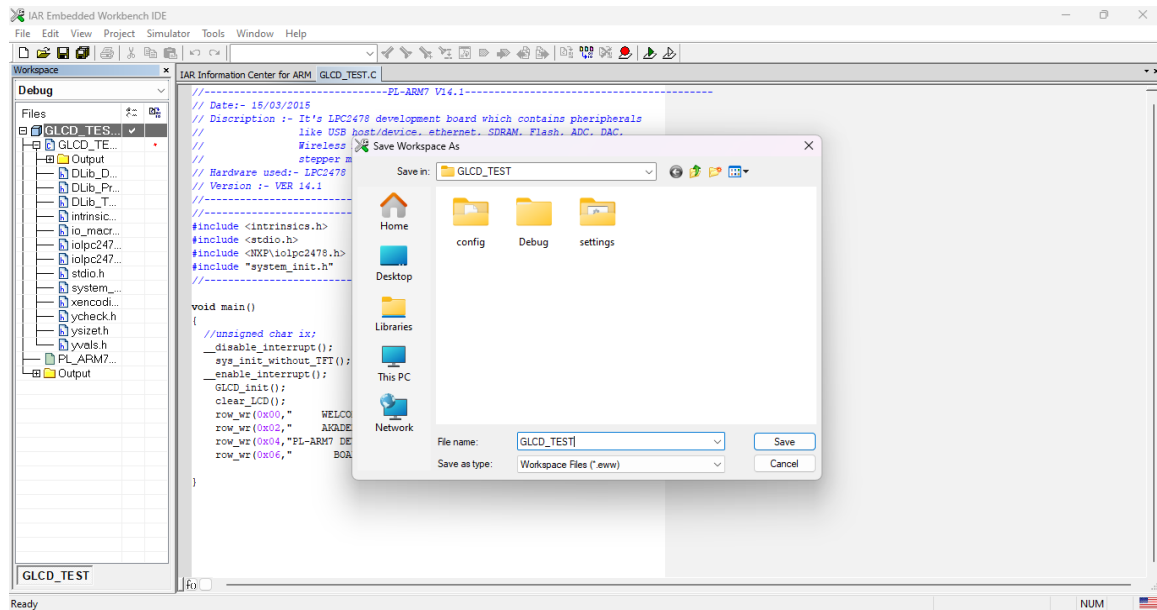
16. Right Click on **GLCD_TEST** ->Add -> Select **Add Files**



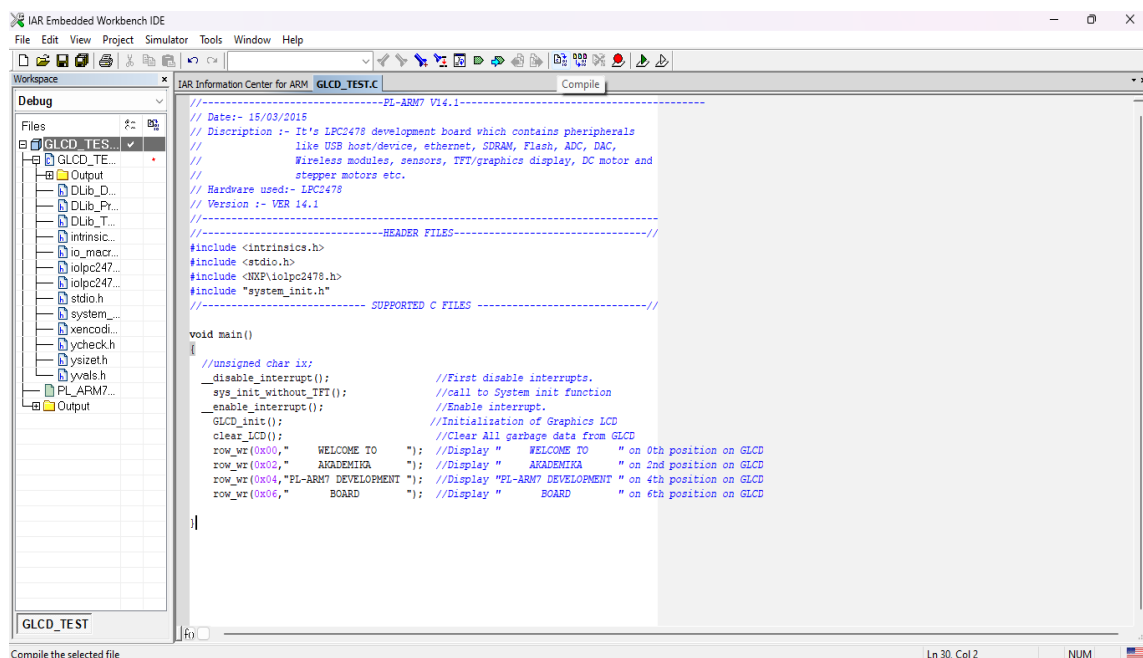
17. File name: **PL_ARM7_TOTAL_LIB** -> Select Files of type: **All Files** -> **Open**



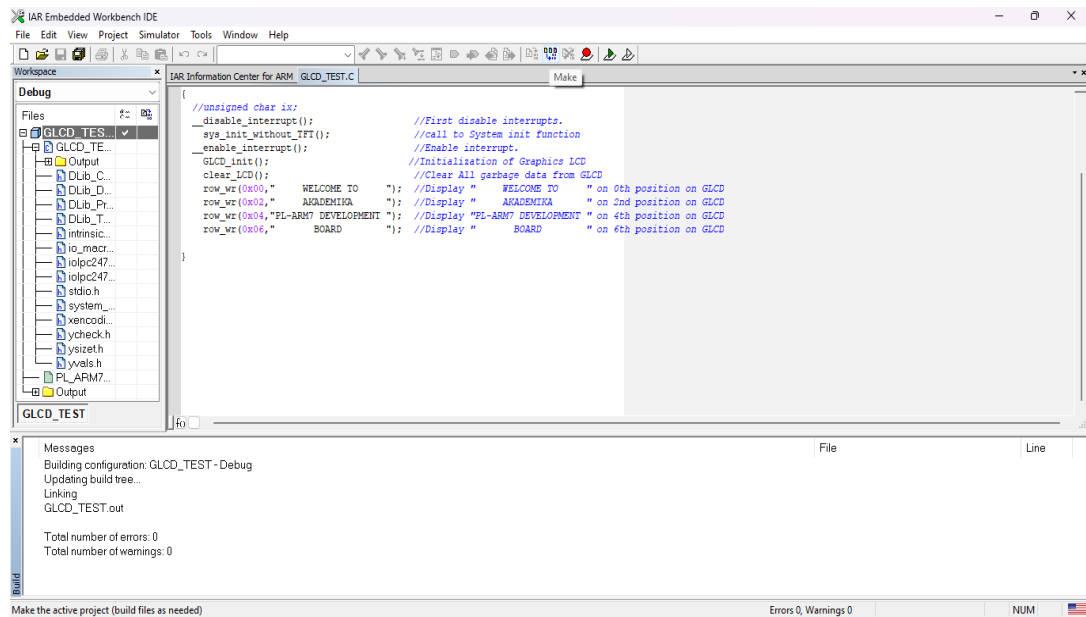
18. Type File name: **GLCD_TEST** -> Select Save type as: **Workspace Files** -> **Save**



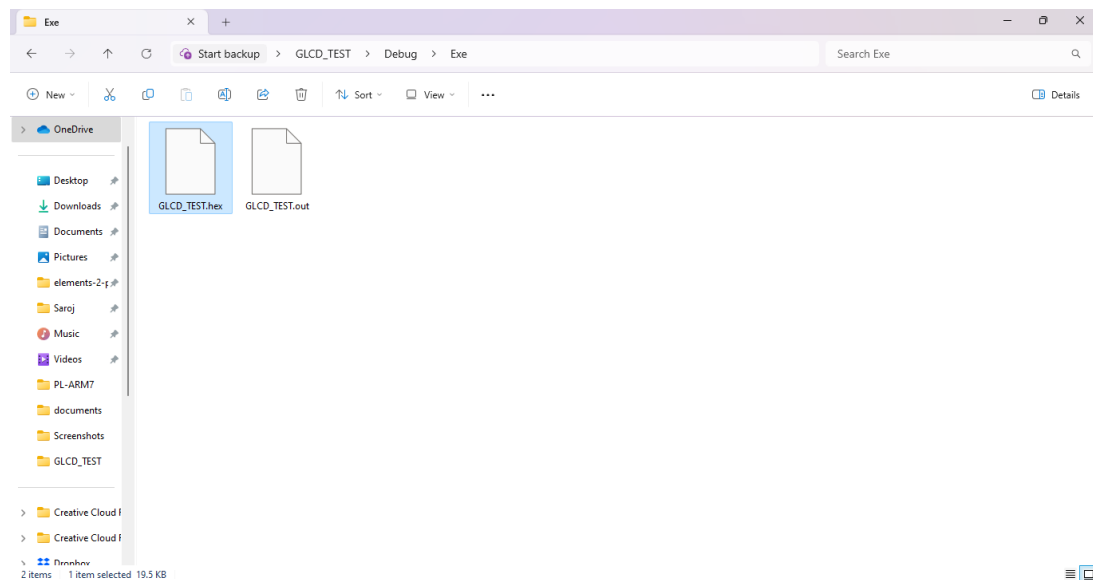
19. click on **Compile** and check program error



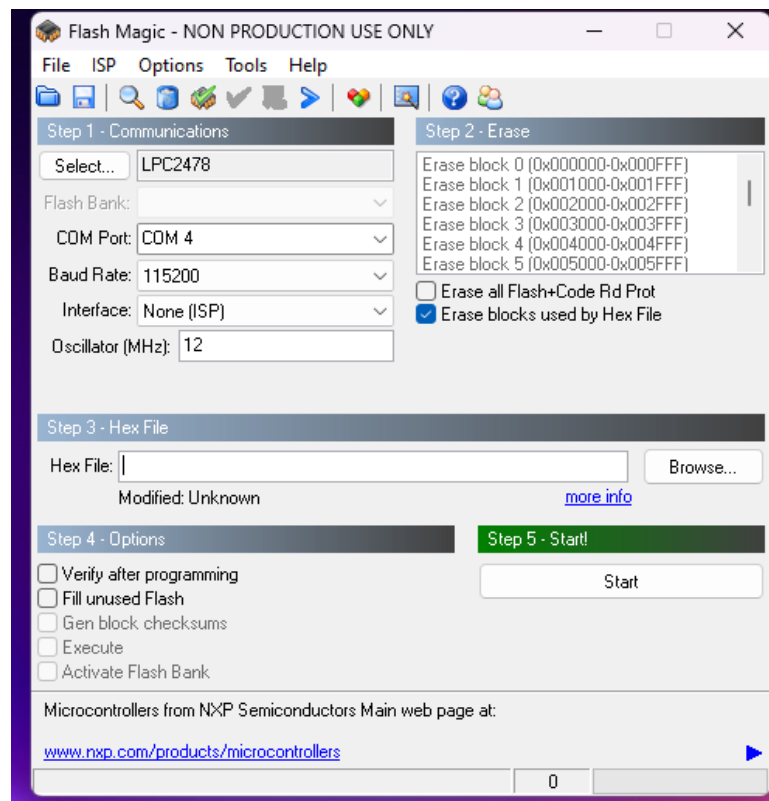
20. Check the program and click on **Make** (make means save the program).

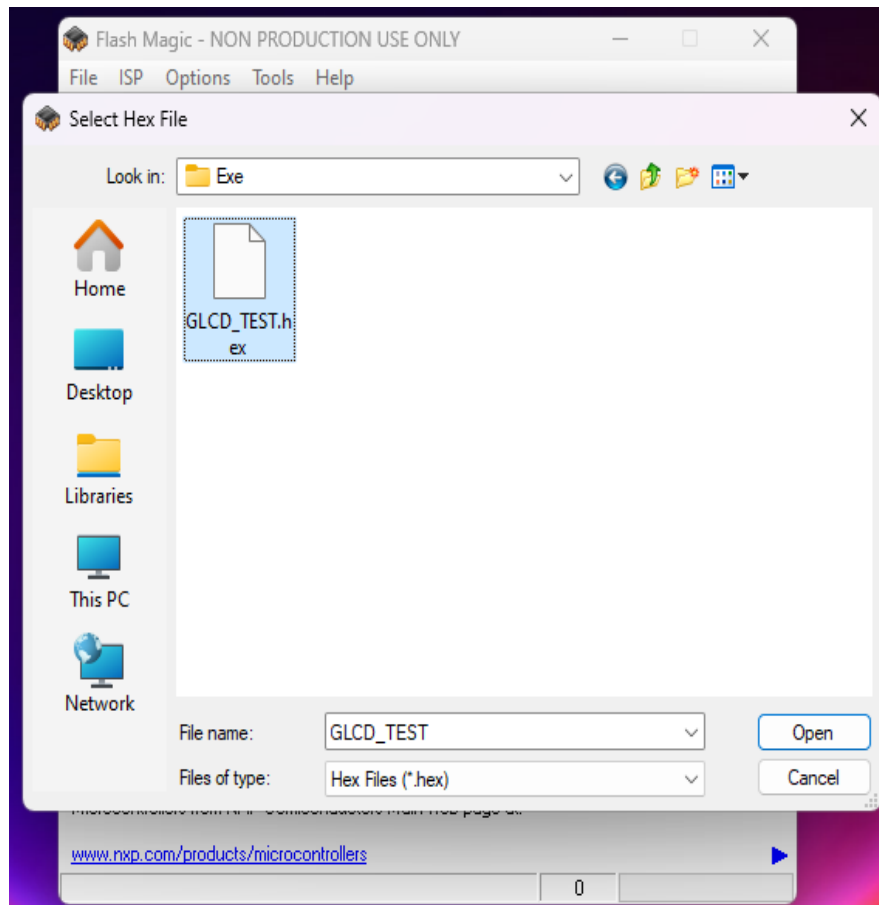


21. Go to **GLCD_TEST ->Release ->Exe -> GLCD_TEST** file has been created



22. Open **Flash magic** Software and select program then click **Start** and check **output** on the **PL-ARM7** Kit.





EXAMPLE NO: -1
GLCD INTERFACE

GLCD interface with LPC2478

AIM:-

Graphics display interface with LPC2478.

OBJECTIVES:-

To study Graphics LCD display interface.

THEORY:-

Graphics LCD display is design to display different types of fonts on display or any image also.

We use 128 x 64 dots GLCD.

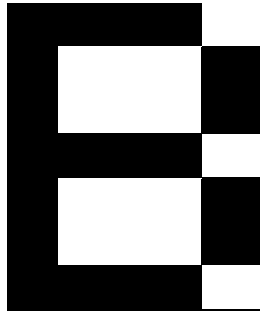
It has 128 column & 64 Rows to display character.

128x64 RAM pattern is as follows.

	C0	C1	C2					C127
R0	D0	D0	D0	.	.	.	.	D0
R1	D1	D1	D1	.	.	.	.	D1
R2	D2	D2	D2	.	.	.	.	D2
R3	D3	D3	D3	.	.	.	.	D3
R4	D4	D4	D4	.	.	.	.	D4
R5	D5	D5	D5	.	.	.	.	D5
R6	D6	D6	D6	.	.	.	.	D6
R7	D7	D7	D7	.	.	.	.	D7
.								
.								
.								
.								
.								
.								
R56	D0	D0	D0	.	.	.	.	D0
R57	D1	D1	D1	.	.	.	.	D1
R58	D2	D2	D2	.	.	.	.	D2
R59	D3	D3	D3	.	.	.	.	D3
R60	D4	D4	D4	.	.	.	.	D4
R61	D5	D5	D5	.	.	.	.	D5
R62	D6	D6	D6	.	.	.	.	D6
R63	D7	D7	D7	.	.	.	.	D7

--	--	--	--	--	--	--	--	--

If anyone wants to display character 'B' then values are as follows



According to above settings we got a value as follows

B = {0x7F, 0X49, 0X49, 0X49, 0X36};

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Download PL-ARM7_GLCD.hex using Flash Magic from following path
**\\DELIVAREABLE\ PROGRAMS)_PL-ARM7_VER
 14.1_GLCD\PROJECT\Release\Exe**
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

				W	E	L	C	O	M	E		T	O					
					A	K	A	D	E	M	I	K	A					
P	L	-	A	R	M	7		D	E	V	E	L	O	P	M	E	N	T
					B	O	A	R	D									

EXAMPLE NO: -2
RELAY/BUZZER
INTERFACE

Relay / Buzzer interface with LPC2478

AIM:-

Relay / Buzzer interface with LPC2478.

OBJECTIVES:-

To study Relay / Buzzer interface.

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Connect 2 pin jumper cap between 1 to 2 for Relay and 2 to 3 for Relay.
- Download PL-ARM7_RELAY_&_BUZZER.hex using Flash Magic from following path
\\PL_ARM7\DELIVAREABLE PROGRAMS) PL-ARM 7_VER 14.1_RELAY_&_BUZZER\PROJECT\Release\Exe
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

				W	E	L	C	O	M	E		T	O						
					A	K	A	D	E	M	I	K	A						
P	L	-	A	R	M	7		D	E	V	E	L	O	P	M	E	N	T	
					B	O	A	R	D										

- Relay or Buzzer will toggle according to program.

EXAMPLE NO: -3
SWITCH & LED
INTERFACE

Switch & LED interface with LPC2478

AIM:-

Switch & LED interface with LPC2478.

OBJECTIVES:-

To study Switch & LED interface.

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Connect 2 pin jumper cap between 2 & 3 of J57.
- Download PL-ARM7_SWITCH_&_LEDS.hex using Flash Magic from following path

\\DELIVAREABLE\ PROGRAMS) PL-

ARM7_VER14.1_SWITCH_&_LEDS\PROJECT\Release\Exe

- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

				W	E	L	C	O	M	E		T	O						
					A	K	A	D	E	M	I	K	A						
P	L	-	A	R	M	7		D	E	V	E	L	O	P	M	E	N	T	
					B	O	A	R	D										

- According to selected switch, corresponding LED will be ON and status of that LED will be display on GLCD, as follows.

				W	E	L	C	O	M	E		T	O					
					A	K	A	D	E	M	I	K	A					
		L	E	D	1		O	F	F									
		L	E	D	2		O	F	F									
		L	E	D	3		O	F	F									
		L	E	D	4		O	F	F									

EXAMPLE NO: -4
ADC
INTERFACE

ADC interface with LPC2478

AIM:-

ADC interface with LPC2478.

OBJECTIVES:-

To study External & internal ADC interface.

THEORY:-

External ADC:-

ADC is the analog to digital convertor.

We use 12bit, 1.25MSPS parallel ADC with internal reference voltage 2.5V for our experiment.

ADC having following control signals, as follows.

1. ADC BUSY :- Busy signal while converting analog data to digital
2. ADC RD# :- This signal is to use read converted data from
ADC
3. ADC CS# :- This signal is user to activate ADC for operation.
4. ADC CONVST# :- Start of conversion pulse.
5. D0 to D11 :- Digital Data.

It is possible to measure conversion time by measuring pulse width of ADC busy signal.

ADC resolution is as follows.

Resolution = Reference voltage / $(2^{12} - 1)$.

Resolution = 2.5V / 4095

Resolution = 0.610mV

Internal ADC:

LPC2478 having 8bit internal ADC

Resolution of ADC is as follows

Resolution = Reference voltage / $(2^{10} - 1)$.

Resolution = 3.3V / 1023

Resolution = 3.22mV

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Connect 16 2pin jumper caps to U23 connector.
- Connect link form J43 to J68 and connect link from J44 to J54 for external ADC
- Connect link form J43 to J53 and connect link from J44 to J69 for internal ADC
- Download PL-ARM 7_ADC_AD7492.hex using Flash Magic from following path for external ADC
**\\DELIVAREABLE\ PROGRAMS) PL-ARM 7_VER
 14.1_ADC7492 \PROJECT\Release\Exe**
- Download PL-ARM7_INTERNAL_ADC.hex using Flash Magic from following path for internal ADC
**\\DELIVAREABLE \PROGRAMS\06)PL-ARM7_VER
 14.1_INTERNAL_ADC\PROJECT\Release\Exe**
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

				W	E	L	C	O	M	E		T	O						
					A	K	A	D	E	M	I	K	A						
P	L	-	A	R	M	7		D	E	V	E	L	O	P	M	E	N	T	
					B	O	A	R	D										

- According to variation of variable pot, ADC value will be shown on GLCD display, as follows.

				W	E	L	C	O	M	E		T	O						
					A	K	A	D	E	M	I	K	A						
P	L	-	A	R	M	7		D	E	V	E	L	O	P	M	E	N	T	
		A	D	C	=	1	.	2	2	2	2	V							

EXAMPLE NO: -5
DAC INTERFACE

DAC interface with LPC2478

AIM:-

ADC interface with LPC2478.

OBJECTIVES:-

To study External & internal ADC interface.

THEORY:-

External DAC:-

DAC is the digital to analog convertor.

We use 12bit serial parallel DAC with internal reference voltage 4.095V for our experiment.

DAC having following control signals, as follows.

1. DAC LD# :- Load serial data to DAC register.
2. DAC CLR# :- Clear DAC register data.
3. DAC CS# :- This signal is user to activate DAC to receive
Serial data.
4. DAC SCK :- SPI clock signal.
5. DAC MOSI :- SPI data in signal.

Resolution of DAC is 1mV/count.

Internal DAC:

LPC2478 having 8bit internal ADC

Reference voltage for DAC is 3.3V

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.

- Connect Jumper caps to J9 & J10 from ISP section.
- Connect DSO probe to J70 and GND to J56 for external DAC.
- Connect DSO probe to J55 and GND to J71 for internal DAC.
- Download PL-ARM 7_DAC8512.hex using Flash Magic from following path for external DAC
\\PL_ARM7\DELIVAREABLE PROGRAMS) PL-ARM 7_VER14.1_DAC8512\PROJECT\Release\Exe
- Download PL-ARM7_INTERNAL_DAC.hex using Flash Magic from following path for internal ADC
\\DELIVAREABLE\ PROGRAMS) PL-ARM 7_VER 14.1_INTERNAL_DAC\PROJECT\Release\Exe
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
						P	L	-	A	R	M	7								
	G	E	N	E	R	A	T	I	O	N		O	F		R	A	M	P		
						O	N		D	A	C									

- Ramp wave form will be generated on J55 or J70.

EXAMPLE NO: -6
SENSOR INTERFACE

Sensor interface with LPC2478

AIM:-

Sensor interface with LPC2478.

OBJECTIVES:-

To study temperature sensor & LDR interface.

THEORY:-

Temperature sensor:-

We are using LM35 as a temperature sensor for our experiment.

It has temperature sensing range from -55°C to 150°C.

LM35 having resolution of 10.0mV/°C.

Reference voltage at 25°C for LM35 is 250mV.

According to that current temperature calculation is as follows.

$TEMP = (((Current\ vtg - 0.2500) / 0.010) + 25);$

LDR

We are using TO-8 photocell as a LDR.

It having Tolerance: ±40% @1 ftc, ±33% @ 2 ftc

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Connect link form J41 to J53 and connect link from J42 to J69 for temperature sensor.
- Connect link form J39 to J53 and connect link from J40 to J69 for LDR

- Download **PL-ARM7_TEMP_SENSOR.hex** using Flash Magic from following path for Temperature sensor
**\\DELIVAREABLE\ PROGRAMS) PL-ARM 7_VER
14.1_TEMP_SENSOR\PROJECT\Release\Exe**
- Download **PL-ARM7_LDR.hex** using Flash Magic from following path for LDR
**\\PL_ARM7\DELIVAREABLE PROGRAMS) PL_ARM7_VER
14.1_LDR \PROJECT\Release\Exe**
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							

- In temperature sensor experiment GLCD display shows current temperature, as follows

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
			T	E	M	P	=	3	1	.	0	*	C							

- In LDR experiment GLCD display shows voltage across LDR, as follows.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
	L	D	R		V	T	G	=	1.	2	2	5	4	V						

EXAMPLE NO: -7
MOTOR INTERFACE

MOTOR interface with LPC2478

AIM:-

MOTOR interface with LPC2478.

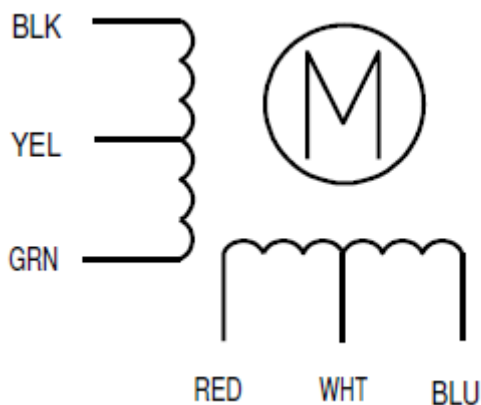
OBJECTIVES:-

To study stepper & DC motor interface.

THEORY:-

Stepper Motor:-

We are using Bipolar Stepper motor which having 1.8° step angle. This stepper motor contain 2 phase, as shown below



BLK & GRN is one phase; RED & BLU is another phase.
YEL & WHT is +12V supply.

We can program this stepper motor in different methods
Half wave driving, Full wave driving.

DC motor

We are using 12V DC motor for our experiment.

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Connect Stepper motor to J14 connector, for stepper motor experiment.
- Connect link between J58 to J46 and J59 to J45 and also connect DC motor to J19.
- Download **PL-ARM 7_STEPPER_MOTOR.hex** using Flash Magic from following path for stepper motor.
\\PL_ARM7\DELIVAREABLE PROGRAMS) PL-ARM 7_VER 14.1_STEPPER_MOTOR\PROJECT\Release\Exe
- Download **PL-ARM7_DC_MOTOR.hex** using Flash Magic from following path for DC motor.
\\DELIVAREABLE\ PROGRAMS) PL-ARM 7_VER14.1_DC_MOTOR\PROJECT\Release\Exe
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							

- After that Stepper motor & DC start rotating.

EXAMPLE NO: -8
KEYPAD INTERFACE

KEYPAD interface with LPC2478

AIM:-

KEYPAD interface with LPC2478.

OBJECTIVES:-

To study keypad interface.

THEORY:-

We use matrix keypad in our experiment.

	SCN0	SCN1	SCN2	SCN3
R0	KEY 0	KEY 4	KEY 8	KEY C
R1	KEY 1	KEY 5	KEY 9	KEY D
R2	KEY 2	KEY 6	KEY A	KEY E
R3	KEY 3	KEY 7	KEY B	KEY F

Scan lines are the output lines and Return lines are the input lines to the processor.

Initially all return lines scan lines are high.

While scanning make SCN0 line low and read return line, if no any key pressed then all return lines are high.

If any key pressed then that particular line is low, so we can detect pressed key.

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.

- Download **PL-ARM7_HEX_KEYPAD**.hex using Flash Magic from following path.
\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER 14.1_HEX_KEYBOARD\PROJECT\Release\Exe
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							

- After that, press any key.
- GLCD will display message for pressed key as follows

		K	E	Y	0		P	R	E	S	S	E	D							

EXAMPLE NO: -9
I2C BASED EEPROM
INTERFACE

I2C BASED EEPROM interface with LPC2478

AIM:-

I2C based EEPROM interface with LPC2478.

OBJECTIVES:-

To study I2C based EEPROM interface.

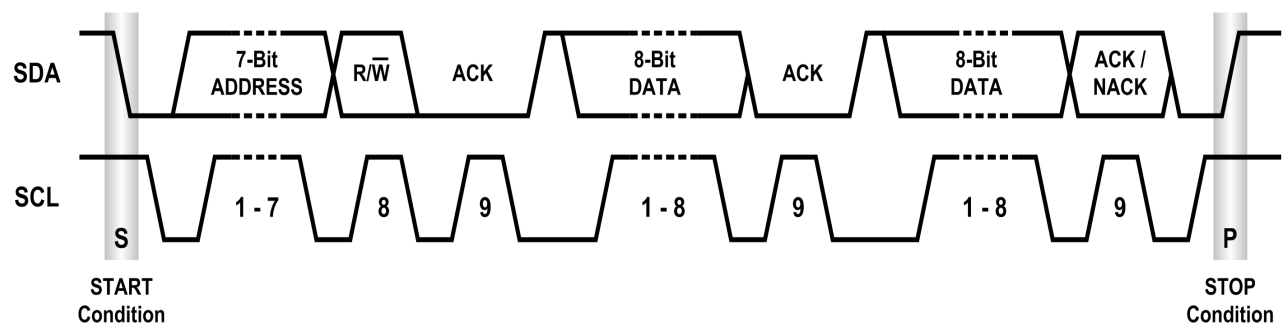
THEORY:-

We use AT24C512 I2C based EEPROM in our experiment.

It having 512KB memory arranges in 8byte boundary.

I2C bus having SDA & SCL line, SCL is clock & SDA is data line.

Following is the frame of I2C.



I2C bus having following format

1. **Start condition:-** Indication of communication start
2. **Slave address:-** 7bit address of interested slave device.
3. **R/W bit:-** Decide read or write operation.
4. **Slave acknowledgement:-** Response from selected slave.
5. **8 bit Data:-** 8 bit data to be read or transfer.
6. **Stop Condition:-** Indication of communication completion.

For AT24C512 we have to send following format

A. EEPROM Write

1. Start Condition

2. Slave address 0x50
3. R/W bit is 0
4. Slave acknowledgement.
5. 8 bit higher address.
6. Slave acknowledgement.
7. 8 bit lower address
8. Slave acknowledgement.
9. 8 bit data
10. Slave acknowledgement.
11. Stop Condition.

B. EEPROM Read

1. Start Condition
2. Slave address 0x50
3. R/W bit is 0
4. Slave acknowledgement.
5. 8 bit higher address.
6. Slave acknowledgement.
7. 8 bit lower address
8. Slave acknowledgement.
9. Repetitive start condition.
10. Slave address 0x50
11. R/W bit is 1.
12. 8 bit data read
13. Give acknowledgement
14. 8 bit data read
15. Give acknowledgement
16. Stop Condition.

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- 1st write data to EEPROM.
- Download PL-ARM 7_EEPROM_ **WRITE**.hex using Flash Magic from following path to write data in EEPROM.

**\\DELIVAREABLE\ PROGRAMS\ 14\PL-ARM 7_VER
14.1_EEPROM_WRITE\PROJECT\Release\Exe**

- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							

- After completion of write EEPROM following message will display.
- " Data Write Completed "**
- Connect Jumper caps to J9 & J10 from ISP section.
 - After completion of writing in EEPROM, Download **PL-ARM7_EEPROM_READ**.hex using Flash Magic from following path.

**\\PL-ARM 7\DELIVAREABLE PROGRAMS\PL-ARM 7_VER
14.1_EEPROM_READ\PROJECT\Release\Exe**

- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, GLCD will display message as follows.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							

- And then GLCD shows EEPROM read data

		D	A	T	A	=	1													

EXAMPLE NO: -10
EXTERNAL INTERRUPT
INTERFACE

External Interrupt interface with LPC2478

AIM:-

External interrupt interface with LPC2478.

OBJECTIVES:-

To External interrupt interface.

THEORY:-

Interrupt is the small service routine which comes suddenly while running main program.

Mainly in microcontroller system we are using timer interrupts, slave devices communication, asynchronous communication.

In timer interrupt, interrupt service routine is call after specified time as per initialization.

In hardware protocol like I2C, SPI, UART etc having master & slave device communication. Master device sends signal to slave which is act as an interrupt to slave.

En LPC2478, it has four external interrupts.

We are using one external interrupt pin from LPC2478.

We are passing square wave to external interrupt. And interrupt is set on rising edge of the square wave.

To observe interrupt behavior will take LED indication.

Normally LED is off, when rising edge of external signal comes then LED gets ON.

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.

- Download **PL-ARM 7_EXTERNAL_INTERRUPT.hex** using Flash Magic from following path.
**\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER
14.1_EXTERNAL_INTERRUPT\PROJECT\Release\Exe**
- Connect square wave to J66 with GND on J67.
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							

- You can vary the frequency of square from 100Hz to 2KHz and observe the speed of blinking of LED.

EXAMPLE NO: -11
FLASH INTERFACE

FLASH interface with LPC2478

AIM:-

FLASH interface with LPC2478.

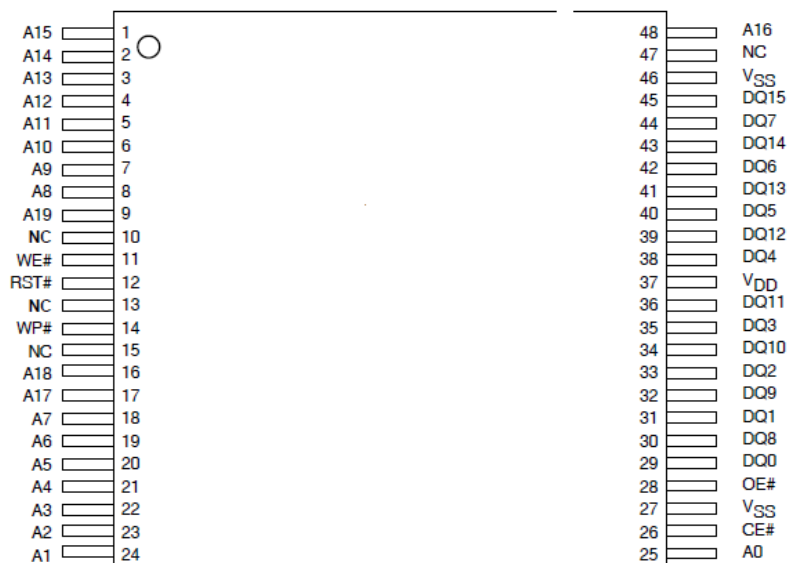
OBJECTIVES:-

To study Flash interface.

THEORY:-

We are using 2MB 16 bit Flash in our experiment.

Following diagram explain the detail pin diagram of Flash.



This flash is 16 bit boundary flash. Each address increment corresponds to 16 bit of parallel data.

It consists of A0 to A19 address lines and D0 to D15 data lines.

LPC2478 have EMC peripheral which controls external static memory and Dynamic memories.

Memory map of LPC2478 is as follows.

The address ranging from 0x80000000 to 0xDFFFFFFF is allocated for external memory interface.

In that,

0x80000000 to 0x83FFFFFF is allocated for 4 static memories with each of 16MB address space

0xA0000000 to 0xDFFFFFFF is allocated for 4 dynamic memories with each of 256MB address space.

Address range	General use	Address range details and description	
0x0000 0000 to 0x3FFF FFFF	On-chip non-volatile memory and Fast I/O	0x0000 0000 - 0x0007 FFFF	Flash Memory (512 kB)
		0x3FFF C000 - 0x3FFF FFFF	Fast GPIO registers
0x4000 0000 to 0x7FFF FFFF	On-chip RAM	0x4000 0000 - 0x4000 FFFF	RAM (64 kB)
		0x7FE0 0000 - 0x7FE0 3FFF	Ethernet RAM (16 kB)
		0x7FD0 0000 - 0x7FD0 3FFF	USB RAM (16 kB)
0x8000 0000 to 0xDFFF FFFF	Off-Chip Memory	Four static memory banks, 16 MB each	
		0x8000 0000 - 0x80FF FFFF	Static memory bank 0
		0x8100 0000 - 0x81FF FFFF	Static memory bank 1
		0x8200 0000 - 0x82FF FFFF	Static memory bank 2
		0x8300 0000 - 0x83FF FFFF	Static memory bank 3
		Four dynamic memory banks, 256 MB each	
		0xA000 0000 - 0xAFFF FFFF	Dynamic memory bank 0
		0xB000 0000 - 0xBFFF FFFF	Dynamic memory bank 1
		0xC000 0000 - 0xCFFF FFFF	Dynamic memory bank 2
		0xD000 0000 - 0xDFFF FFFF	Dynamic memory bank 3
0xE000 0000 to 0xEFFF FFFF	APB Peripherals	36 peripheral blocks, 16 kB each	
0xF000 0000 to 0xFFFF FFFF	AHB peripherals		

This flash is command based flash, It consists different command for write data, read unique no of flash & erase data. Only read operation is perform by giving address and reading data lines.

Following table explains the flash commands.

Command Sequence	1st Bus Write Cycle		2nd Bus Write Cycle		3rd Bus Write Cycle		4th Bus Write Cycle		5th Bus Write Cycle		6th Bus Write Cycle	
	Addr ¹	Data ²	Addr ¹	Data ²	Addr ¹	Data ²	Addr ¹	Data ²	Addr ¹	Data ²	Addr ¹	Data ²
Word-Program	5555H	AAH	2AAAH	55H	5555H	A0H	WA ³	Data				
Sector-Erase	5555H	AAH	2AAAH	55H	5555H	80H	5555H	AAH	2AAAH	55H	SA _x ⁴	30H
Block-Erase	5555H	AAH	2AAAH	55H	5555H	80H	5555H	AAH	2AAAH	55H	BA _x ⁴	50H
Chip-Erase	5555H	AAH	2AAAH	55H	5555H	80H	5555H	AAH	2AAAH	55H	5555H	10H
Erase-Suspend	XXXXH	B0H										
Erase-Resume	XXXXH	30H										
Query Sec ID ⁵	5555H	AAH	2AAAH	55H	5555H	88H						
User Security ID Word-Program	5555H	AAH	2AAAH	55H	5555H	A5H	WA ⁶	Data				
User Security ID Program Lock-Out	5555H	AAH	2AAAH	55H	5555H	85H	XXH ⁶	0000H				
Software ID Entry ^{7,8}	5555H	AAH	2AAAH	55H	5555H	90H						
CFI Query Entry	5555H	AAH	2AAAH	55H	5555H	98H						
Software ID Exit ^{9,10} /CFI Exit/Sec ID Exit	5555H	AAH	2AAAH	55H	5555H	F0H						
Software ID Exit ^{9,10} /CFI Exit/Sec ID Exit	XXH	F0H										

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Download **PL-ARM 7_FLASH_WRITE**.hex using Flash Magic from following path for writing a flash.

\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER

14.1_FLASH_WRITE\PROJECT\Release\Exe

- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
		P	R	E	S	S		S	W	2		T	O		E	R	A	S	E	

- Then press SW2 from keypad, then entire flash will get erased and following message will display on GLCD.

F	L	A	S	H		E	R	A	S	E	C	O	M	P	L	E	T	E	D
	P	R	E	S	S		S	W	3		T	O		W	R	I	T	E	

- Then press SW3 from keypad, the program will write Flash_data[] array data to flash. And write completed message will display on GLCD.
- Connect Jumper caps to J9 & J10 from ISP section.
- After that, to read, download **PL-ARM 7_FLASH_READ.hex** using Flash Magic from following path for reading flash.
**\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER
14.1_FLASH_READ\PROJECT\Release\Exe**
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
	P	R	E	S	S		S	W	3		T	O		R	E	A	D			

- Then press SW3 for keypad, program will read data 512 no of data from 0x00000000 address.

EXAMPLE NO: -12
USB HOST INTERFACE

USB HOST interface with LPC2478

AIM:-

USB HOST interface with LPC2478.

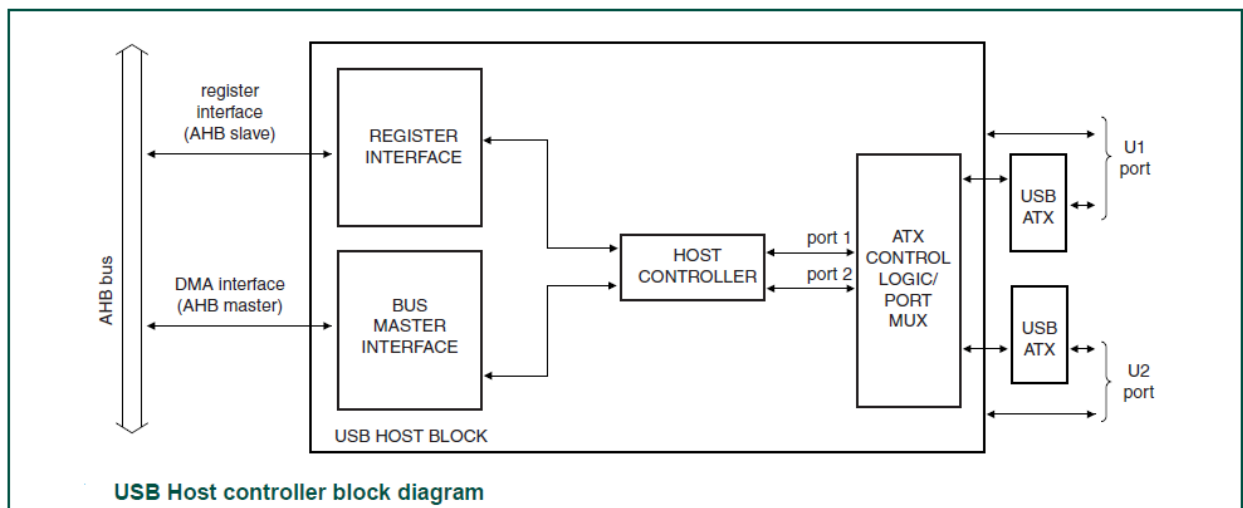
OBJECTIVES:-

To study USB HOST interface.

THEORY:-

LPC2478 supports USB Host and USB device.

USB HOST controller block diagram is as follows



USB HOST means LPC2478 act as a computer.

In this experiment, we implemented USB mass storage class.

PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.

- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Create MSREAD.txt file in any pen drive and write any character in that file.
- Download **UsbHostLite**.hex using Flash Magic from following path.
**\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER
14.1_USB_HOST\PROJECT\Test22_USBHOST\usb.host.lite\
UsbHostLite-Beta-1.0\Obj**
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, connect pen drive to J48 from USB HOST section.
- Program will detect connected pen drive and create new MSWRITE.txt file and copy MSREAD.txt file into that file.

EXAMPLE NO: -13
USB DEVICE INTERFACE

USB DEVICE interface with LPC2478

AIM:-

USB DEVICE interface with LPC2478.

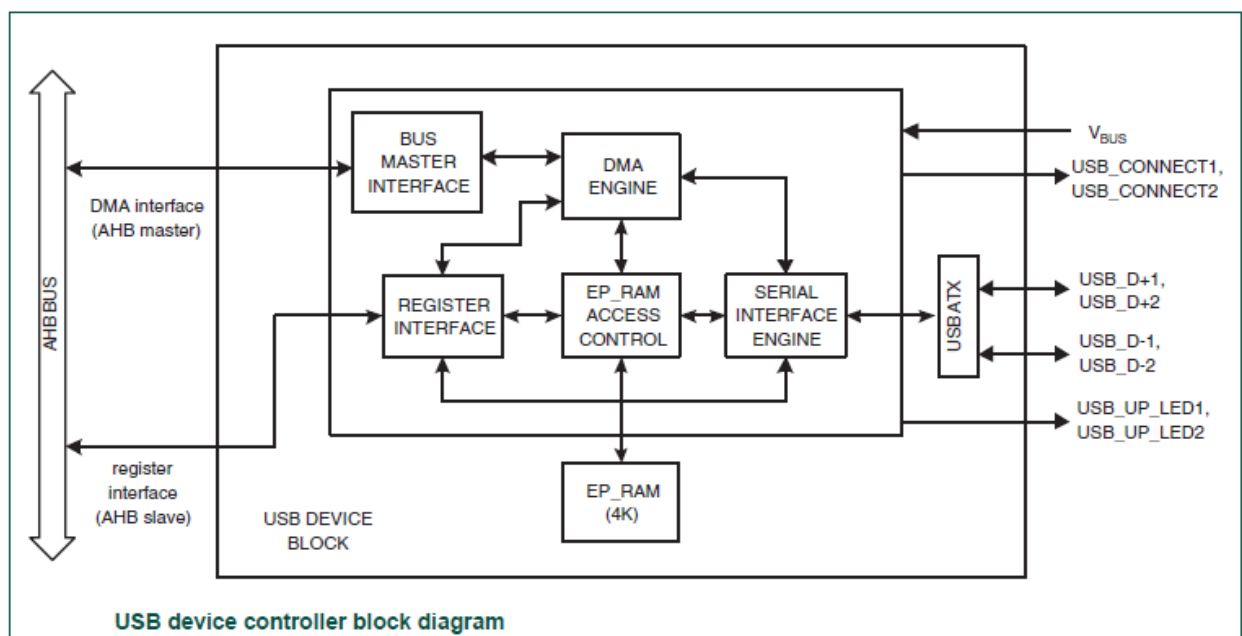
OBJECTIVES:-

To study USB DEVICE interface.

THEORY:-

LPC2478 supports USB Host and USB device.

USB HOST controller block diagram is as follows



USB DEVICE means LPC2478 act as a slave device like pen drive. In this experiment, we implemented USB mass storage class.

PROCEDURE: -

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Download **Memory**.hex using Flash Magic from following path.
**\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER
14.1_USB_DEVICE\PROJECT\Test20_USBMem\Obj**
- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, connect USB cable to J49 from USB DEVICE section.
- PC will ask for drive.
- Then click on automatic drive installation. It will install mass storage USB driver.
- After completion of driver installation LPC2478 act as a pen drive in My Computer
- It contain ReadMe.txt file.

EXAMPLE NO: -14
CAN BUS INTERFACE

CAN BUS interface with LPC2478

AIM:-

CAN BUS interface with LPC2478.

OBJECTIVES:-

To study CAN BUS interface.

THEORY:-

Controller Area Network (CAN) is the definition of a high performance communication protocol for serial data communication. The CAN Controller is designed to provide a full implementation of the CAN-Protocol according to the CAN Specification Version 2.0B.

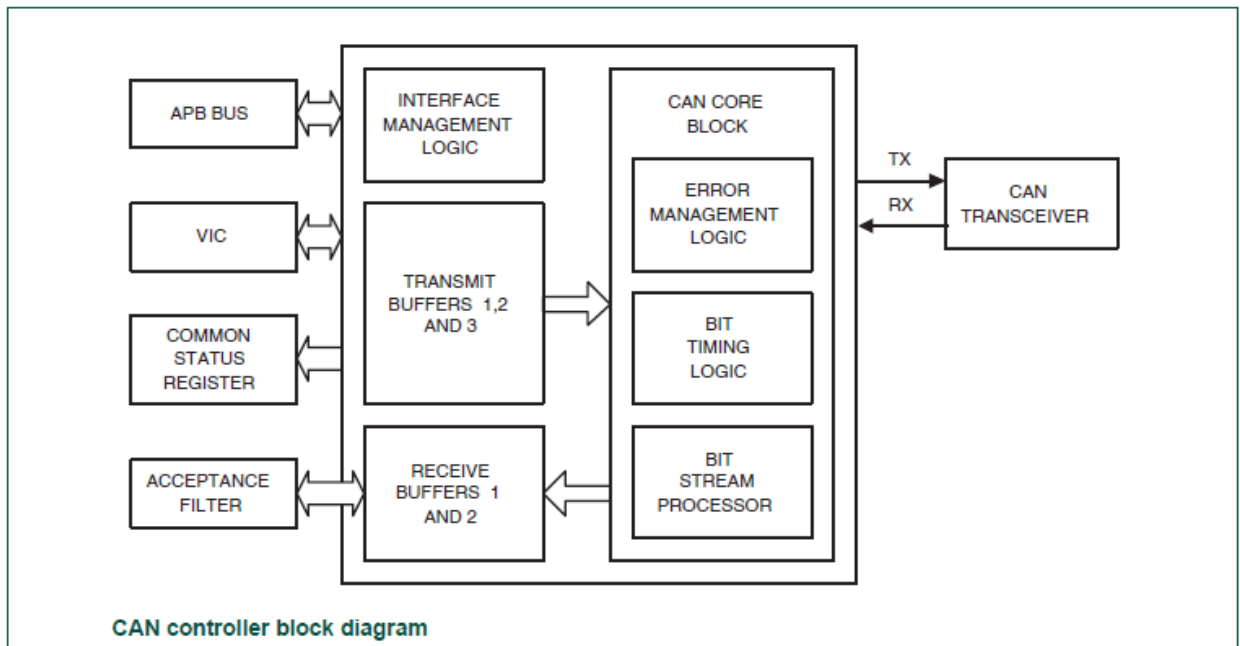
Microcontrollers with this on-chip CAN controller are used to build powerful local networks by supporting distributed real-time control with a very high level of security. The applications are automotive, industrial environments, and high speed networks as well a slow cost multiplex wiring. The result is a strongly reduced wiring harness and enhanced diagnostic and supervisory capabilities.

The CAN block is intended to support multiple CAN buses simultaneously, allowing the device to be used as a gateway, switch, or router among a number of CAN buses in various applications.

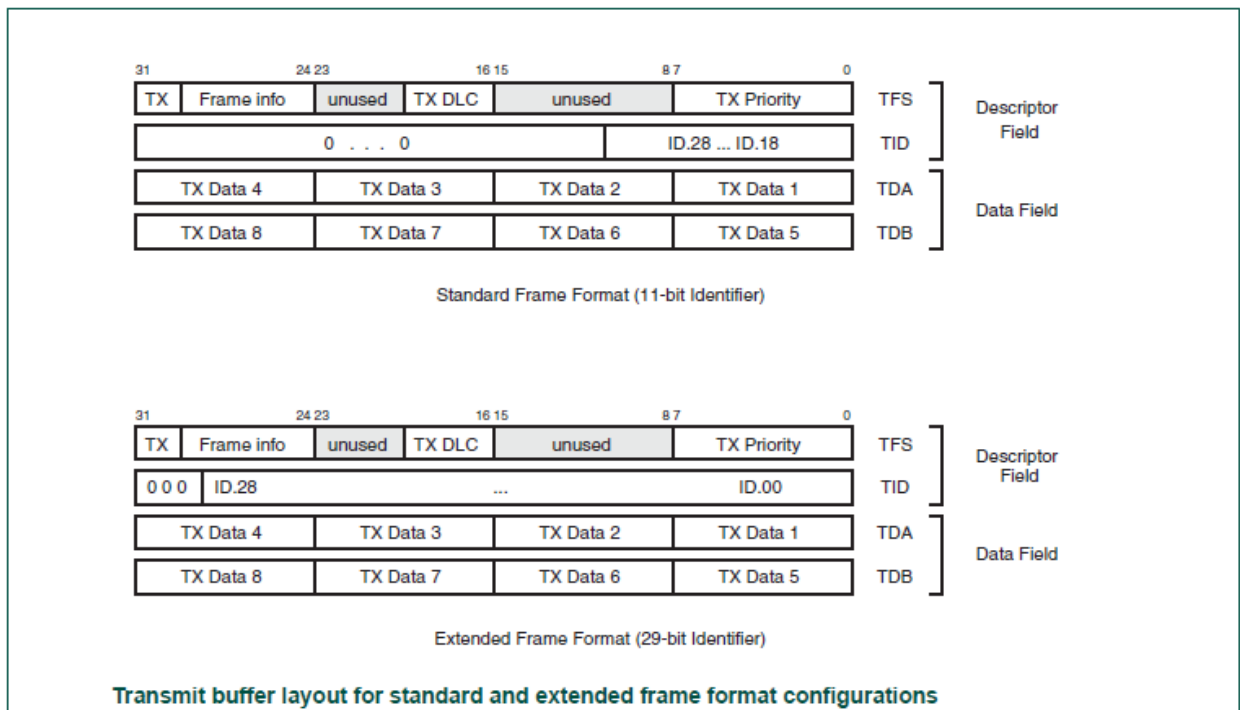
General CAN features

- Compatible with *CAN specification 2.0B, ISO 11898-1*.
- Multi-master architecture with nondestructive bit-wise arbitration.
- Bus access priority determined by the message identifier (11-bit or 29-bit).
- Guaranteed latency time for high priority messages.
- Programmable transfer rate (up to 1 Mbit/s).
- Multicast and broadcast message facility.
- Data length from 0 up to 8 bytes.
- Powerful error handling capability.
- Non-return-to-zero (NRZ) coding/decoding with bit stuffing.

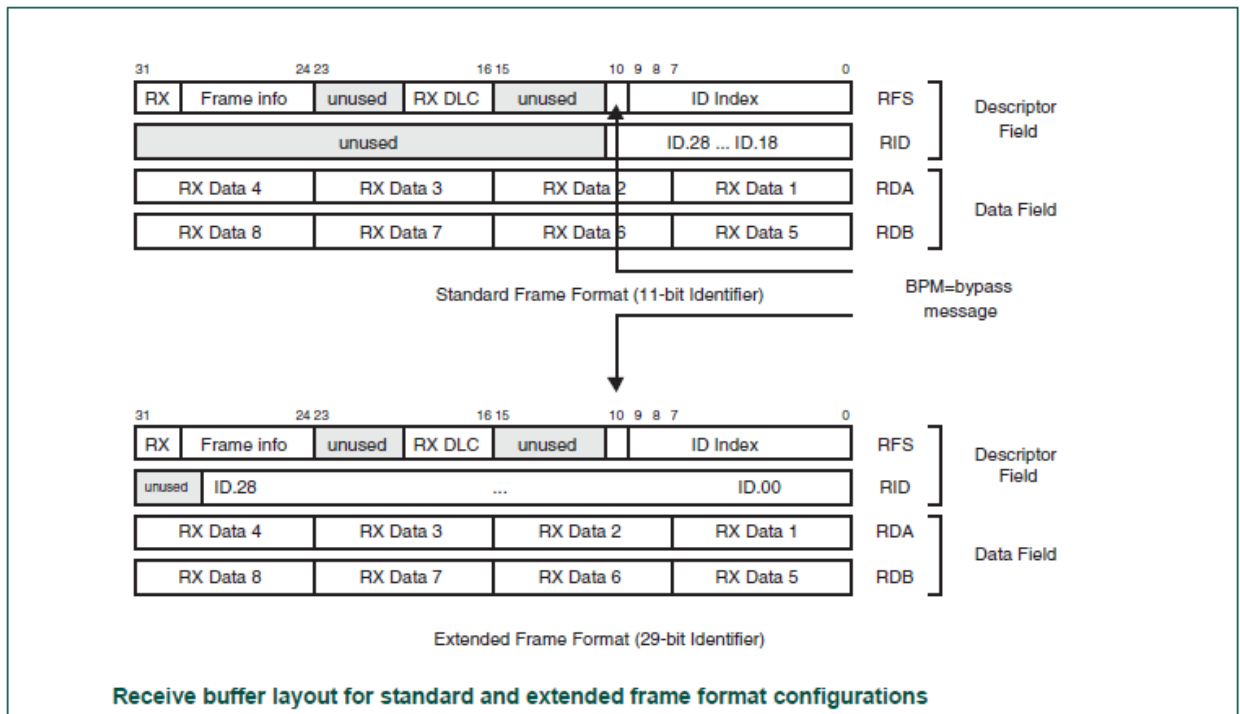
CAN controller block diagram



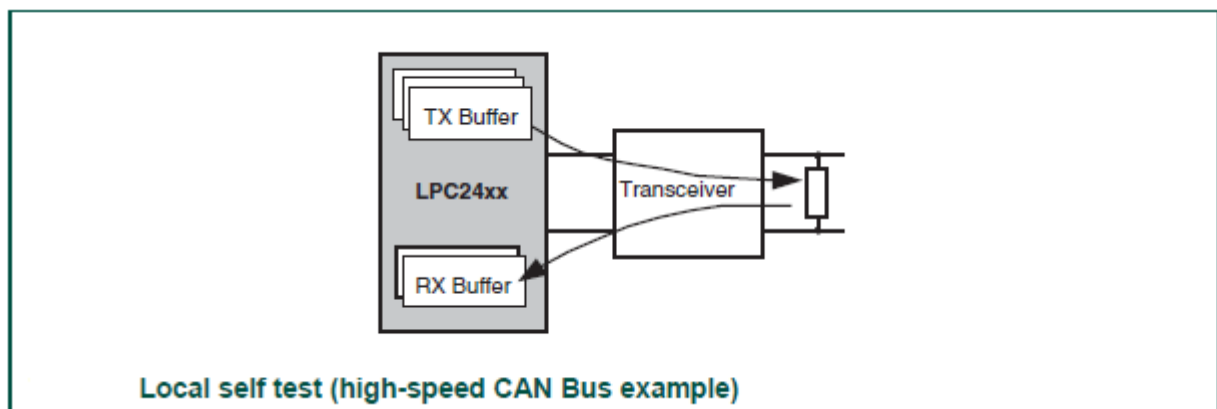
Transmit Buffers (TXB)



Receive Buffer (RXB)



Local self test



PROCEDURE: -

- Make sure that GLCD is connected to J8 in LCD DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Download **PL-ARM 7_CAN_SELF_TEST**.hex using Flash Magic from following path for **self test**.

**\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER
14.1_CAN_SELF_TEST\PROJECT\Release\Exe**

- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
	P	R	E	S	S		S	W	2											

- Then press SW2 from keypad, then program will send created data frame to TX and received on RX line write & following message display on GLCD.

C	A	N		B	U	S		I	N		S	E	L	F		T	E	S	T
	T	X		I	D			=	2	0									
	T	X	D	A	T	A		A	=	5	5	A	A	5	5	A	A		
	T	X	D	A	T	A		B	=	5	5	A	A	5	5	A	A		
	R	X		I	D			=	2	0									
	R	X	D	A	T	A		A	=	5	5	A	A	5	5	A	A		
	R	X	D	A	T	A		B	=	5	5	A	A	5	5	A	A		

- Download **PL-ARM 7_CAN_TX**.hex using Flash Magic from following path for **CAN TX** in one board.

\\DELIVAREABLE\ PROGRAMS) PL-ARM7_VER

14.1_CAN_TX\PROJECT\Release\Exe

- Download **PL-ARM 7_CAN_RX.hex** using Flash Magic from following path for **CAN RX** in one board.

\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER

14.1_CAN_RX\PROJECT\Release\Exe

- Connect 2 flying cables for TX to RX board in J52 CAN BUS section.
- After completion, following message will display on GLCD form **TX** board.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
			C	A	N		D	E	M	O										
	P	R	E	S	S		S	W	2											

- Then press SW2 for keypad, it will initialize can drive for transmitter & following message will display on GLCD.

C	A	N		B	U	S		I	N		T	X		M	O	D	E		
	T	X		I	D				=	2	0								
	T	X	D	A	T	A		A	=	4	4	3	3	2	2	1	1		
	T	X	D	A	T	A		B	=	8	8	7	7	6	6	5	5		
	P	R	E	S	S		S	W	3										

- Then press SW3 for keypad, then GLCD shows TX running message.

	T	X		R	U	N	N	I	N	G									
--	---	---	--	---	---	---	---	---	---	---	--	--	--	--	--	--	--	--	--

- In RX board, GLCD shows following message.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
	C	A	N		D	E	M	O												
	P	R	E	S	S		S	W	2											

- Then press SW2 for keypad, it will initialize can drive for receiver & following message will display on GLCD.

P	R	E	S	S		S	W	3		T	O		R	E	C	E	I	V	E
---	---	---	---	---	--	---	---	---	--	---	---	--	---	---	---	---	---	---	---

- Then press SW3 from keypad, it will receive transmitted data, and give acknowledgment to transmitter, and received data will be displayed on GLCD.

	C	A	N		B	U	S		I	N		R	X		M	O	D	E	
		R	X		I	D		=		2	0								
	R	X		D	A	T	A		A	=		4	4	3	3	2	2	1	1
					R	X	D												
	A	T	A		B	=			8	8	7	7	6	6	5	5			

- After receiving acknowledgment TX board shows TX completed message on GLCD

EXAMPLE NO: -15
ETHERNET INTERFACE

ETHERNET interface with LPC2478

AIM:-

Ethernet interface with LPC2478.

OBJECTIVES:-

To study Ethernet interface.

THEORY:-

The Ethernet block contains a full featured 10 Mbps or 100 Mbps Ethernet MAC (Media Access Controller) designed to provide optimized performance through the use of DMA hardware acceleration. Features include a generous suite of control registers, half or full duplex operation, flow control, control frames, hardware acceleration for transmit retry receive packet filtering and wake-up on LAN activity. Automatic frame transmission and reception with Scatter-Gather DMA off-loads many operations from the CPU.

The Ethernet block and the CPU share a dedicated AHB subsystem (AHB2) that is used to access the Ethernet SRAM for Ethernet data, control, and status information. All other AHB traffic in the LPC2400 takes place on a different AHB subsystem, effectively separating Ethernet activity from the rest of the system. The Ethernet DMA can also access off-chip memory via the External Memory Controller, as well as the SRAM located on AHB1, if is not being used by the USB block. However, using memory other than the Ethernet SRAM, especially off-chip memory, will slow Ethernet access to memory and increase the loading of AHB1.

The Ethernet block interfaces between an off-chip Ethernet PHY using the MII (Media Independent Interface) or RMII (reduced MII) protocol and the on-chip MIIM (Media Independent Interface Management) serial bus.

Ethernet standards support:

- Supports 10 or 100 Mbps PHY devices including 10 Base-T, 100 Base-TX, 100 Base-FX, and 100 Base-T4.
- Fully compliant with IEEE standard 802.3.
- Fully compliant with 802.3x Full Duplex Flow Control and Half Duplex backpressure.
- Flexible transmit and receive frame options.
- VLAN frame support.

Memory management:

- Independent transmits and receives buffers memory mapped to share SRAM.
- DMA managers with scatter/gather DMA and arrays of frame descriptors.
- Memory traffic optimized by buffering and pre-fetching.

Enhanced Ethernet features:

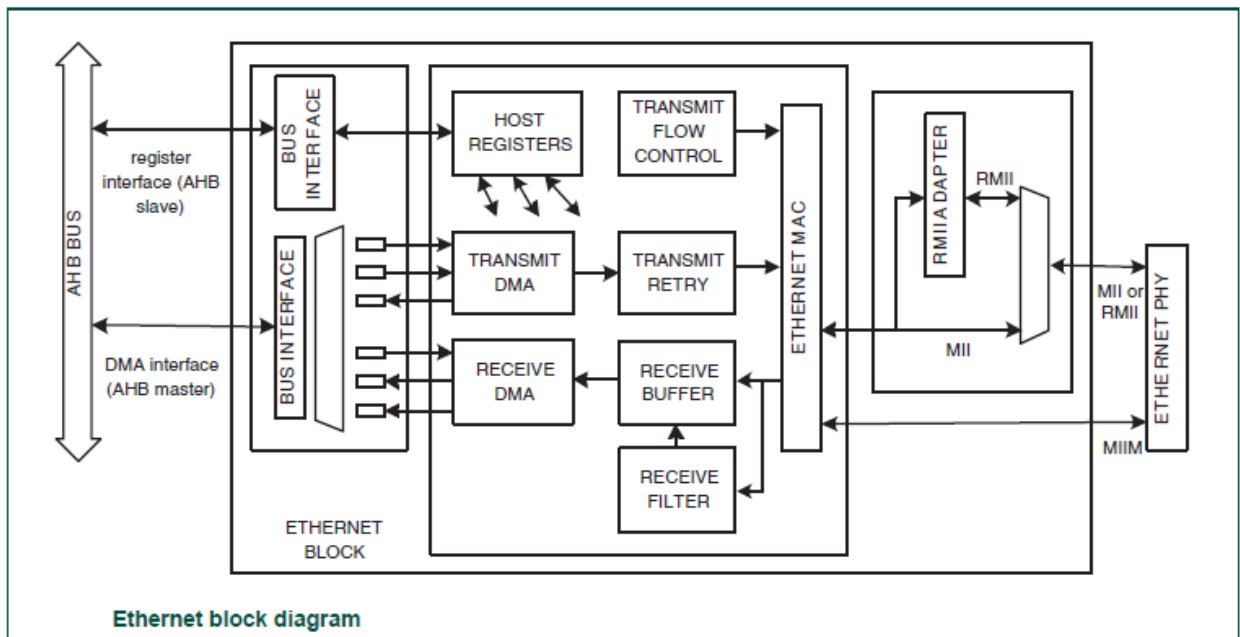
- Receive filtering.
- Multicast and broadcast frame supports for both transmit and receive.
- Optional automatic FCS insertion (CRC) for transmit.
- Selectable automatic transmit frame padding.
- Over-length frame support for both transmit and receive allows any length frames.
- Promiscuous receive mode.
- Automatic collision back off and frame retransmission.
- Includes power management by clock switching.
- Wake-on-LAN power management support allows system wake-up: using the receive filters or a magic frame detection filter.

Physical interface:

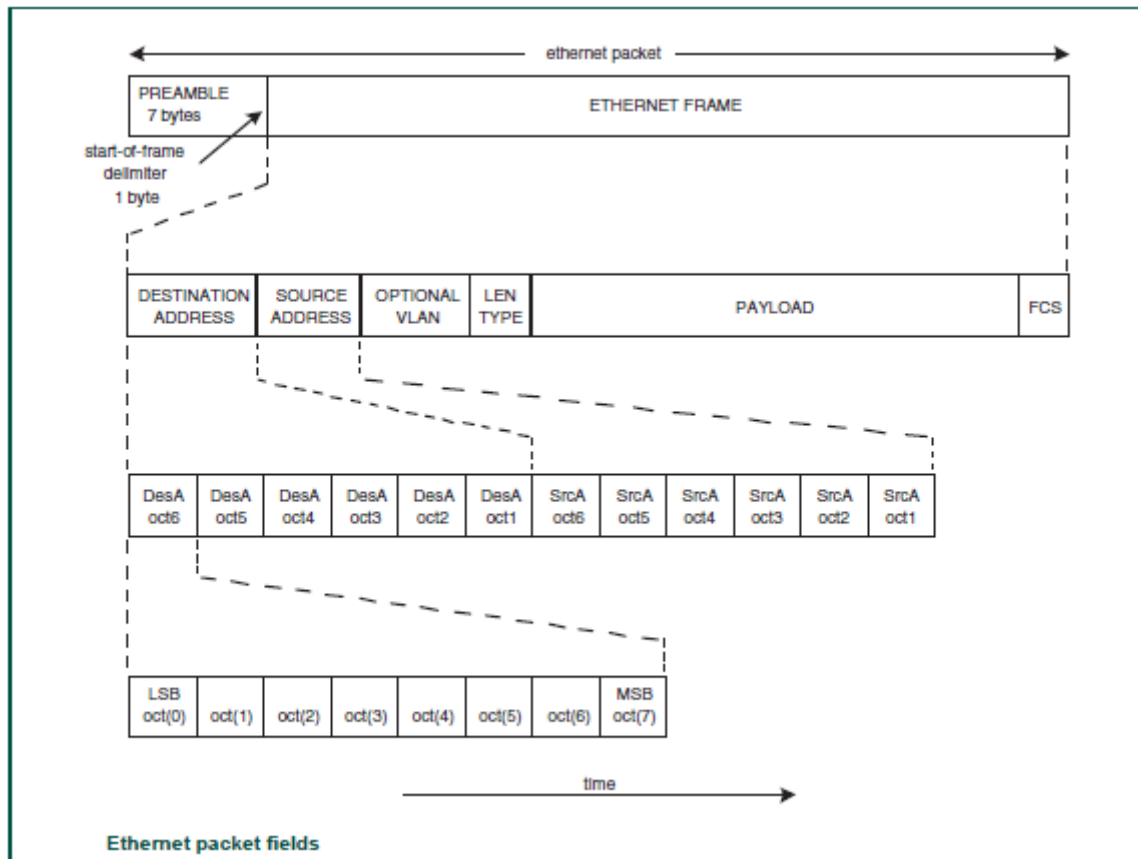
- Attachment of external PHY chip through standard Media Independent Interface (MII) or standard Reduced MII (RMII) interface, software selectable.

- PHY register access is available via the Media Independent Interface Management(MIIM) interface.

Ethernet controller block diagram



Ethernet packet fields



PROCEDURE:-

- Make sure that GLCD is connected to J8 in LCD/TFT DISPLAY section.
- Make sure that Jumper caps are connected to J63 & 64 at 2→3 position.
- Connect ISP USB to PC.
- Connect Jumper caps to J9 & J10 from ISP section.
- Download **PL-ARM 7_ETHERNET_TX.hex** using Flash Magic from following path for **TX board**.
\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER 14.1_ETHERNET_TX\PROJECT\Release\Exe
- Download **PL-ARM 7_ETHERNET_TX.hex** using Flash Magic from following path for **RX board**.

\\DELIVAREABLE\ PROGRAMS\PL-ARM 7_VER 14.1_ETHERNET_TX\PROJECT\Release\Exe

- Remove Jumper caps to J9 & J10 from ISP section and restart board.
- After completion, following message will display on GLCD for TX & RX board.

	W	E	L	C	O	M	E		T	O		A	K	A	D	E	M	I	K	A
							P	L	-	A	R	M	7							
	P	R	E	S	S		S	W	2											

- Then press SW2 from keypad, both Ethernet PHY get reset.
- After reset again press sw2 key in TX board, wait for some time & press SW2 key in RX board. GLCD will display all register value of PHY.
- Then press SW6 form TX board, it will generate Ethernet frame and send to RX board.
- Open HyperTerminal from RX board connected PC.
- Press SW6 from RX board, it will display transmitted message.